

A reformulation of the theory of unimolecular reactions

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The theory of unimolecular reactions is reformulated in a way which makes no reference to the concepts of a transition state, a reaction coordinate, or of an activated complex: the present theory is based, instead, on a simple master-equation approach using the estimated positions and lifetimes of the rotation-vibration energy levels of the molecule. The fundamental basis of this reformulation is that the unimolecular rate is derived from the perturbation of the normal modes of internal relaxation of the reactant molecule by the microscopic reaction processes. The thermal decompositions of cyclobutane, cyclopropane, ethyl isocyanide, ethyl chloride, and methyl isocyanide are examined in detail, and it is shown that not only is the present theory much easier to use than is conventional RRKM theory, but also that it gives better results.

An Appendix provides a set of algorithms required for these calculations.

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On propose une reformulation de la théorie des réactions unimoléculaires d'une façon telle que l'on ne fasse aucune référence aux concepts d'état de transition, de coordonnée de réaction ou de complexe activé: on base plutôt la nouvelle théorie sur une équation maîtresse simple utilisant des positions et des temps de vie évalués des niveaux énergétiques rotation-vibration de la molécule. Le concept fondamental de cette reformulation repose sur la notion que la vitesse unimoléculaire provient d'une perturbation des modes normaux de relaxation interne de la molécule réactive par des processus réactionnels microscopiques. On a examiné en détail les décompositions thermiques du cyclobutane, du cyclopropane, de l'isocyanure d'éthyle, du chlorure d'éthyle et d'isocyanure de méthyle et on a montré que la théorie est plus facile à utiliser que la théorie RRKM conventionnelle et qu'elle donne aussi de meilleurs résultats.

Un appendice fournit l'ensemble d'algorithmes requis pour ces calculs.

[Traduit par le journal]

I. Introduction

This examination of the theory of unimolecular reactions was undertaken principally out of a desire to see the theory cast in modern molecular-dynamic terms. We do not claim that the theory set out below satisfies this desire completely: in a purely formal sense it does, but in practice our estimates of the effective lifetimes of the reactive states with respect to predissociation are not, at present, derived by molecular-dynamic methods.

The most successful current theory of thermal unimolecular reactions is the RRKM theory (1-6), but despite its obvious success in describing reasonably well a very wide range of reaction processes, it suffers from some frustrating deficiencies. Most importantly, although much emphasis is placed upon the selection of a hypothetical reaction coordinate and the corresponding activated complex, it hardly matters what reaction coordinate is chosen (5, 7, 8) because the calculated fall-off behaviour is very insensitive to this choice. Less serious, but nevertheless unsettling, are the problems surrounding the assumed collisional deactivation efficiency: one is that markedly different collision efficiencies are often required to make the calculated and experimental fall-off curves coincide at different temperatures (5), or sometimes these efficiencies do not vary monotonically

with temperature (9, 10); another is that collision efficiencies of the order of unity are often found, and yet it is reasonably well established that the vessel walls are more effective in transferring internal energy than are collisions with other molecules (11-13). These difficulties do not arise in the present reformulation: the concepts such as reaction coordinate do not appear, and it becomes evident in the later discussion (section XIB) that the puzzle surrounding collision efficiencies is part experimental in origin, and part a matter of theoretical interpretation.

II. List of Symbols

The nomenclature used in this presentation has been chosen to remain, as far as possible, in conformity with existing practices (5) where an identical or nearly identical concept is being represented; for the remainder, we have chosen the simplest nomenclature which appears possible.

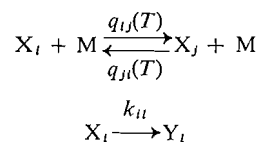
- A internal relaxation matrix (s^{-1}); elements a_{ij} .
- A_{∞} Arrhenius frequency factor at infinite pressure (s^{-1}).
- a_i coefficient in the numerator of the generalised Lindemann form, eqs. 46, 47; $\bar{a}_i$ is grouped value in graining procedure, and $\bar{\bar{a}}_i$ is scaled value used in the practical algorithm.

- B** symmetrised relaxation matrix (s^{-1}).
 b_i i th coefficient in the denominator of the generalised Lindemann form, eqs. 46, 47; $\bar{b}_i$ and $\tilde{b}_i$ are respectively grouped and scaled quantities.
 β inverse temperature (cm^{-1} ; $= 1/kT$).
D predissociation matrix; diagonal elements $d_i = -k_i$ (s^{-1}).
E equilibrium population matrix—non-dissipative system; diagonal elements $\bar{n}_j$.
 ϵ energy (cm^{-1} or kcal mol^{-1}).
 ϵ_0 critical energy (cm^{-1} or kcal mol^{-1} ; $= \epsilon_\infty$ assuming eq. 34).
 ϵ_∞ activation energy at infinite pressure for strict Arrhenius rate law.
 ϵ_{act} activation energy at arbitrary pressure.
 g_i degeneracy of state i .
 $G(\epsilon)$ state-sum up to energy ϵ .
 Γ diagonalised reaction matrix; γ_j is j th eigenvalue.
 k Boltzmann constant.
 k_{il} predissociation rate constant from reactant in state i to product in state l (s^{-1}).
 k_i decay rate constant from state i to all product channels (s^{-1} ; $= \sum_l k_{il}$).
 $k(\epsilon)$ average decay rate constant over all states i at energy ϵ (s^{-1}).
 k_{uni} unimolecular rate constant (s^{-1} ; $= -\gamma_{n-1}$).
 k_∞ infinite pressure rate constant (s^{-1}).
 $\mathcal{L}$ Laplace transform.
 Λ diagonalised relaxation matrix; λ_j is j th eigenvalue.
 λ collisional efficiency in RRKM theory.
 $\bar{\lambda}$ collisional efficiency or pressure-shift parameter.
[M] concentration of inert-gas molecules (molecules cm^{-3}).
 n number of levels in relaxing system.
 $\bar{n}_i$ equilibrium (fractional) population of the i th state—non-dissipative system; $\sum_i \bar{n}_i = 1$.
 $\bar{n}_r$ total equilibrium (fractional) population within the r th reactive channel grain.
 $\eta_i(t)$ population of the i th state at time t —dissipative system; $\sum_i \eta_i(0) = 1$.
 p pressure (Torr).
 p_{ij} transition probability per collision from state i to state j .
 $P(\epsilon)$ state-degeneracy at energy ϵ ; $= \sum_i g_i$ over states at ϵ .
 $\rho(\epsilon)$ density of states at energy ϵ (cm^{-1}).
 q_{ij} transition rate constant from state i to state j ($\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$).
R symmetrised reaction matrix (s^{-1}).
S eigenvector matrix of relaxation matrix **B**; S_{ij} is i th element of j th eigenvector.

- Ψ** eigenvector matrix of reaction matrix **R**; ψ_{ij} is i th element of j th eigenvector.
 s Kassel shape parameter for unimolecular fall-off curve.
 σ collision diameter.
 T temperature.
 μ internal relaxation time constant of molecule (s^{-1}).
 μ' internal relaxation rate constant ($\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$).
 $\bar{\mu}$ internal relaxation rate constant ($\text{Torr}^{-1} \text{ s}^{-1}$; $= \mu' \times 9.88 \times 10^{18} T^{-1}$).
 ν_i vibrational frequency of i th oscillator (cm^{-1}).
 x_i anharmonicity factor of vibration, assumed to be a Morse oscillator.
 $Z(T)$ collision number at temperature T ($\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$).
 $\mathcal{Q}(\beta)$ partition function at inverse temperature β .

III. The Reaction Model

This treatment follows from an extension of the primitive master-equation model we described earlier this year (14). The reactant molecule X is considered to possess a denumerable set of rotation-vibration states X_i , and is diluted in a heat bath of structureless inert gas molecules M; transitions among the various internal states X_i are induced by collisions with M, and should a molecule X find itself in a state with energy greater than a critical amount ϵ_0 , predissociation to products may occur with a set of characteristic time constants, in competition with collisional deactivation. The model then, is no more than a generalised Lindemann scheme (5, 6, 14)



where Y_l represents the product in the l th internal state; for the time being, we ignore the internal structure of the product molecules.

Implicit in the model are the following assumptions. The temperature of the heat bath remains unchanged during the reaction, and the heat-bath molecules maintain their translational equilibrium. The mutual interactions between the X molecules are neglected in order to preserve the linearity of the master equation. The model is also dissipative in the sense that the reverse reaction is excluded, a simplification justified by the fact that most unimolecular reactions are studied far away from equilibrium (14).

In this treatment, we have readopted the natural ordering of subscripts, so that $q_{ij}(T)$ is the rate constant for the transition of X from internal state i to

internal state j upon collision with M at temperature T . This rate constant is related to the transition probability per collision $p_{ij}(T)$ via the collision number $Z(T)$, assumed to be independent of i , viz.

$$[1] \quad q_{ij}(T) = Z(T) \cdot p_{ij}(T)$$

where

$$Z(T) = 2\sigma^2(2\pi/\beta m)^{1/2}$$

σ being the collision diameter and $\beta = 1/kT$, the inverse temperature, and m is the reduced mass of X and M . The quantities q_{ij} and q_{ji} are related by the principle of detailed balancing

$$[2] \quad \tilde{n}_i q_{ij} = \tilde{n}_j q_{ji}$$

where $\tilde{n}_i$ is the equilibrium fractional population of the i th state in the *absence of reaction* (i.e. when $k_{il} = 0$ for all i); the internal-relaxation process is also Markovian, so that

$$[3] \quad \sum_j p_{ij} = 1$$

The microscopic rate constant k_{il} for the conversion of the reactant in state i to the product in state l is assumed to be independent of both temperature and the concentrations of X and M . Thus, the master equation for the evolution of the population of the i th state of X is readily written as

$$[4] \quad \frac{d\eta_i(t)}{dt} = \sum_j a_{ij}\eta_j(t) + d_i\eta_i(t)$$

where $\eta_i(t)$ is the fractional population of the i th state at time t , and

$$[5a] \quad a_{ij} = Z[M]\{(1 - \delta_{ij})p_{ji} - \delta_{ij} \sum_k (1 - \delta_{ik})p_{ik}\}$$

$$[5b] \quad d_i = - \sum_l k_{il}$$

Thus, d_i is zero for all unreactive states, and the summation in [5b] is understood to be over all accessible product channels; the quantity $-d_i$ will be redesignated as k_i later, in section V.

Since [4] is linear, it may be written in matrix notation

$$[6] \quad \frac{d\eta(t)}{dt} = (A + D)\eta(t)$$

where

$$d_{ij} = d_i\delta_{ij}$$

and the matrix sum $(A + D)$ can be symmetrised in the usual way (15) using the equilibrium population vector, i.e.

$$[7] \quad R = E^{-1/2}(A + D)E^{1/2}$$

where

$$e_{ij} = \tilde{n}_i\delta_{ij}$$

and R is now symmetric; under this transformation [6] becomes

$$[8a] \quad \frac{d\chi(t)}{dt} = R\chi(t)$$

with

$$[8b] \quad \chi(t) = E^{-1/2}\eta(t)$$

as the 'symmetrised population evolution vector'. Finally, if R is diagonalised using the matrices Ψ and Ψ^T to give

$$[9] \quad \Gamma = \Psi^T R \Psi$$

Γ has the property that all the eigenvalues are real and negative, i.e.

$$0 > \gamma_{n-1} \gg \gamma_{n-2} \geq \gamma_{n-3} \geq \dots \geq \gamma_0$$

since R is symmetric and negative definite. Because R and $(A + D)$ are related by a similarity transformation, their eigenvalues are identical, and hence (14) we identify the experimentally observed unimolecular rate constant with the eigenvalue of largest algebraic magnitude, i.e.

$$[10] \quad k_{uni} = -\gamma_{n-1}$$

It follows also that the evolution of the populations of the states of the molecule X is given by

$$[11] \quad \eta_i(t) = \tilde{n}_i^{1/2} \sum_j \exp(\gamma_j t) \psi_{ij} \sum_k \psi_{kj} \eta_k(0) \tilde{n}_k^{-1/2}$$

This equation has been used repeatedly in describing non-dissipative systems (16), but it is equally valid for dissipative systems.

The first aim of the master-equation formulation of unimolecular reactions is therefore the determination of the eigenvalue γ_{n-1} ; subsequent aims will involve the determination of the eigenvector elements ψ_{ij} . However, whilst the formal solutions may be readily written, the explicit determination of the exact analytic solution for a general system, with a general set of transition probabilities p_{ij} and microscopic rate constants k_{il} , is a formidable task because of a general lack of theorems in this area; hence, our previous approach using numerical experiment (14). Consequently, we will attempt here to establish, as a first step in this direction, explicit relations between k_{uni} and the normal modes of internal relaxation (17), given a set of microscopic rate constants k_{il} . Note here that the term *internal relaxation* refers to changes between rotation-vibration states of X brought about by collision with M ; thus, this avenue of attack is simply one way of trying to solve the generalised Lindemann scheme, as presented at the beginning of this section.

IV. The Relation Between k_{uni} and the Normal Modes of Relaxation

Following our earlier work on molecular relaxation (16, 17) we define, for the internal relaxation system, the symmetrised transition matrix **B** as

$$[12] \quad \mathbf{B} = \mathbf{E}^{-1/2} \mathbf{A} \mathbf{E}^{1/2}$$

which has eigenvalues λ_j and eigenvectors $\mathbf{S}_j$;

$$[13] \quad \mathbf{B} = \mathbf{S} \mathbf{A} \mathbf{S}^T$$

Recalling briefly the properties of the conservative relaxation system, we have

$$[14] \quad n_i(t) = \tilde{n}_i^{1/2} \sum_j \exp(\gamma_j t) S_{ij} \sum_k S_{kj} n_k(0) \tilde{n}_k^{-1/2}$$

where $n_i(t)$ is the fractional population of state i in the absence of the reaction; notice the similarity with eq. 11. In the conservative case, however, the eigenvalue sequence is

$$0 = \lambda_{n-1} \gg \lambda_{n-2} \geq \lambda_{n-3} \geq \dots \geq \lambda_0$$

with the zero eigenvalue ensuring conservation of particles, and the particular sequence of eigenvector elements

$$[15] \quad S_{i,n-1} = \tilde{n}_i^{1/2}$$

ensuring the attainment of the Boltzmann equilibrium distribution as $t \rightarrow \infty$. Finally

$$[16] \quad \lambda_i = \lambda_i' [\mathbf{M}]$$

where the λ_i' are constants, in other words, the relaxation time constants are proportional to $[\mathbf{M}]$.

We now rewrite the eigenvalue equation [9], i.e.

$$[17] \quad \mathbf{R} \mathbf{\Psi}_i = \gamma_i \mathbf{I} \mathbf{\Psi}_i$$

in the iterative form (18)

$$[17a] \quad \mathbf{\Psi}_i = (\gamma_i \mathbf{I} - \mathbf{B})^{-1} \mathbf{D} \mathbf{\Psi}_i$$

Then, using [13], we have

$$[18] \quad (\gamma_i \mathbf{I} - \mathbf{B})^{-1} = \{\mathbf{S}(\gamma_i \mathbf{I} - \mathbf{A}) \mathbf{S}^T\}^{-1} \\ = \mathbf{S}(\gamma_i \mathbf{I} - \mathbf{A})^{-1} \mathbf{S}^T$$

whence

$$[19] \quad \mathbf{\Psi}_i = \mathbf{S}(\gamma_i \mathbf{I} - \mathbf{A})^{-1} \mathbf{S}^T \mathbf{D} \mathbf{\Psi}_i$$

The perturbed eigenvector elements are then

$$[20] \quad \psi_{ki} = \sum_l \left(\sum_m \frac{S_{km} S_{lm}}{\gamma_i - \lambda_m} \right) \left(\sum_j d_{lj} \psi_{ji} \right)$$

and since **D** is diagonal, this reduces to

$$[20a] \quad \psi_{ki} = \sum_l \sum_m \frac{S_{km} S_{lm}}{\gamma_i - \lambda_m} d_l \psi_{li}$$

remembering that the summation over l refers to the predissociating states only. For a system with r predissociating states [20] leads to a system of r equations in r unknowns, γ_i and the $(r-1)$ ratios of ψ_{ki} . The utility of [20] follows from the particular spacings of the eigenvalues γ_i and λ_i , where, using a theorem of Courant and Hilbert (19), it may be shown that

$$[21] \quad 0 = \lambda_{n-1} < |\gamma_{n-1}| \ll |\lambda_{n-2}| \leq |\lambda_{n-3}| \\ \leq \dots \leq |\lambda_0|$$

whence, to an excellent degree of approximation

$$\psi_{k,n-1} = \sum_l \frac{S_{k,n-1} S_{l,n-1}}{\gamma_{n-1}} \cdot d_l \psi_{l,n-1} \\ - \sum_l \sum_{m=0}^{n-2} \frac{S_{km} S_{lm}}{\lambda_m} \cdot d_l \psi_{l,n-1}$$

or, using [15]

$$[22] \quad \psi_{k,n-1} = \tilde{n}_k^{1/2} \gamma_{n-1}^{-1} \sum_l \tilde{n}_l^{1/2} d_l \psi_{l,n-1} \\ - \sum_l \sum_{m=0}^{n-2} \frac{S_{km} S_{lm}}{\lambda_m} \cdot d_l \psi_{l,n-1}$$

Equation 22 lends itself to further development along two distinct lines. First, it provides a framework for the explicit examination of the dependence of k_{uni} on the normal modes of relaxation, at least for model systems, and will be used in this way in subsequent papers to treat other thermal reactions including dielectronic ionisation of hydrogen atoms, and shock wave dissociation of H_2 and D_2 . Second, its linearity in γ_{n-1}^{-1} facilitates direct evaluation of k_{uni} for realistic systems of known microscopic rate constants k_{il} and normal modes of relaxation. We will pursue the first point insofar as it pertains to the theory of unimolecular reactions in the remainder of this section, and thereafter devote our attention to the second one.

Suppose there is only one predissociating state r , i.e.

$$d_i = d_r \delta_{ir}$$

Then, for the element $\psi_{r,n-1}$, [22] reduces to

$$[23] \quad \psi_{r,n-1} = \left\{ \tilde{n}_r d_r \gamma_{n-1}^{-1} - \sum_{m=0}^{n-2} \frac{S_{rm}^2}{\lambda_m} \cdot d_r \right\} \psi_{r,n-1}$$

which, together with [10] and [16], leads to k_{uni} in the Lindemann form

$$[24] \quad k_{\text{uni}} = -\gamma_{n-1} = \frac{-\tilde{n}_r d_r}{1 - d_r \kappa_{rr} / [\mathbf{M}]}$$

In this equation the term in the denominator

$$[25] \quad \kappa_{rr} = - \sum_{m=0}^{n-2} \frac{S_{rm}^2}{\lambda_m'}$$

represents an average of all the normal modes of relaxation (S_{rm}^2) associated with the state r , weighted by their respective characteristic times $\tau_m = -\lambda_m^{-1}$; in fact, the quantity $\kappa_{rr}/[M] \simeq \tau$, the macroscopic relaxation time for the internal energy. Thus, for a system with only one predissociating state, the internal relaxation amongst its various states has the overall effect of grouping all the nonpredissociating states into one, and the fall-off of k_{uni} with pressure is determined principally by the internal energy re-

laxation rate and, to a lesser extent, by the fine structure of the relaxation, i.e. the detailed nature of the transition probabilities.

It is important to realise that [24] depends on the approximations made in [21], but since

$$|\gamma_{n-1}| \leq \tilde{n}_r |\lambda_{n-2}| \leq \dots \leq \tilde{n}_r |\lambda_0|$$

and $\tilde{n}_r < 10^{-10}$ for systems of interest, the reason for our earlier numerical finding (14) that such a reaction obeyed Lindemann behaviour to better than one part in 10^6 becomes apparent. On the other hand, for a system with two predissociating states labelled r and s , we have, from [22]

$$[26] \quad k_{\text{uni}} = \frac{(\tilde{n}_r \kappa_{rr} + \tilde{n}_s \kappa_{ss} - 2\tilde{n}_r^{1/2} \tilde{n}_s^{1/2} \kappa_{rs}) d_r d_s [M]^{-1} - (\tilde{n}_r d_r + \tilde{n}_s d_s)}{1 + \{(\kappa_{rr} \kappa_{ss} - \kappa_{rs}^2) d_r d_s [M]^{-2} - (\kappa_{rr} d_r + \kappa_{ss} d_s) [M]^{-1}\}}$$

where, as before,

$$[26a] \quad \kappa_{rs} = - \sum_{m=0}^{n-2} \frac{S_{rm} S_{sm}}{\lambda_m'}$$

Thus, with two predissociating channels open, k_{uni} no longer has the Lindemann form, as we found numerically earlier (14); moreover, it can be seen that, in general, k_{uni} cannot be written as a superposition of the Lindemann expressions for the respective channels. This latter observation, then, cuts away at one of the fundamental building blocks of conventional formulations, that k_{uni} can be expressed as the integral over all energy of the differential contribution, dk_{uni} , to the unimolecular rate constant, i.e.

$$[27] \quad k_{\text{uni}} = \int_{\epsilon_0}^{\infty} dk_{\text{uni}}$$

where the dk_{uni} are all in Lindemann form (5). The inadequacy of this integral formulation lies in its failure to take account of the perturbation of one predissociating state by all the other predissociating states: this mutual perturbation is of course implicit in the determination of γ_{n-1} through eq. 9. On the other hand, the failure of this integral representation is not catastrophic, for otherwise the development of theories of unimolecular reactions could not have progressed as far as they have done over the last 50 years; nevertheless, the abandonment of [27] is one of the factors contributing to the superiority of the present reformulation over existing theories.

V. The Choice of the k_i

The use of the master-equation formulation to describe chemically reacting systems presents us with a curious paradox. On the one hand, it would appear

that, even for quite simple molecules, many millions or billions of state-to-state and state-to-product processes are taking place and therefore their rate constants need to be known. On the other hand, it has been found repeatedly for diatomic dissociation reactions that the rates of observable processes are either only weakly dependent on, or else totally independent of, the assumed state-to-state or state-to-product rate constants (20); we have elaborated on this point elsewhere already (14, 20). In this section, therefore, we will deduce an approximate choice for the numerical values of the k_i , in the following section we will suggest a logical choice for the numerical values of the a_{ij} and, thereafter, we will show that these two choices lead to an excellent representation of all the known unimolecular reaction characteristics for five reactions which have been subjected to extensive experimental scrutiny.

At infinite pressure, the rate constant k_{uni} becomes (14)

$$[28] \quad k_{\infty}(\beta) = \sum \tilde{n}_i k_i$$

$$[28a] \quad = \sum k_i g_i \exp(-\beta \epsilon_i) / \mathcal{Q}(\beta)$$

where $\mathcal{Q}(\beta)$ is the molecular partition function of the reactant at the inverse temperature β . Thus, the sum of $g_i k_i$ over all the states i at energy ϵ is the inverse Laplace transform of $k_{\infty}(\beta) \mathcal{Q}(\beta)$; i.e.

$$[29] \quad \sum_{\epsilon} k_i g_i = \mathcal{L}^{-1}\{k_{\infty}(\beta) \mathcal{Q}(\beta)\}$$

where the summation is over states at energy ϵ . We now define $k(\epsilon)$ as the average of k_i over the states at ϵ , i.e.

$$[30] \quad k(\epsilon) = \sum_{\epsilon} k_i g_i / P(\epsilon); \quad P(\epsilon) \neq 0$$

where

$$[31] \quad P(\epsilon) = \sum_i g_i$$

is the state-degeneracy at energy ϵ . Since $\mathcal{Q}(\beta)$ is just the Laplace transform of $P(\epsilon)$, the product $k_\infty(\beta)\mathcal{Q}(\beta)$ in [29] is simply the Laplace transform of the convolution of $\mathcal{L}^{-1}\{k_\infty(\beta)\}$ and $P(\epsilon)$; whence

$$[32] \quad P(\epsilon)k(\epsilon) = \mathcal{L}^{-1}\{k_\infty(\beta)\} * P|_\epsilon$$

where $f * g|_\epsilon$ denotes the convolution of functions f and g from zero to ϵ . Thus, given $k_\infty(\beta)$ one may, at least in principle, obtain $k(\epsilon)$, as was first done by Slater (21) in 1955. Over the years, considerable misgivings have been voiced about the validity of such procedures (22–26). However, it may be seen that the derivation of [32] from [28] is rigorous.¹ In particular, it may be shown that for a *discrete system* of predissociating levels, the $k(\epsilon)$ function is given rigorously by

$$[33] \quad k(\epsilon) = A_\infty P(\epsilon - \epsilon_\infty)/P(\epsilon); \quad P(\epsilon) \neq 0$$

if the rate law is strict Arrhenius in form, i.e.

$$[34] \quad k_\infty(T) \equiv k_\infty(\beta) = A_\infty \exp(-\epsilon_\infty/kT) \\ = A_\infty \exp(-\beta\epsilon_\infty)$$

Similarly, it follows, again rigorously, that if the rate law is in modified Arrhenius form, viz.

$$[35] \quad k_\infty(T) \equiv k_\infty(\beta) = B_\infty T \exp(-\epsilon_\infty^*/kT) \\ = B' \beta^{-1} \exp(-\beta\epsilon_\infty^*)$$

where $B' = B_\infty/k$, then

$$[36] \quad k(\epsilon) = B'G(\epsilon - \epsilon_\infty^*)/\rho(\epsilon)$$

where $G(\epsilon)$ is the sum of states up to and including ϵ and $\rho(\epsilon)$ is the state density at ϵ .

It is customary to present experimental data for the infinite-pressure rate constant k_∞ in Arrhenius form, eq. 34. Whether or not experimental rate data for thermal unimolecular reactions obey [34] *rigorously* is a matter which we will discuss further below, cf. Appendix 1; deviations from [34], on the other hand, are clearly small, so that we may write a generalised form of the Arrhenius equation, i.e.

$$[37] \quad k_\infty = \sum_j A_{\infty j} \exp(-\beta\epsilon_{\infty j}), \quad j = 1, 2, 3, \dots$$

¹It should be remembered, however, that $k_\infty(\beta)$ may not be defined in the form of eq. 28 if β is too small, but our treatment using the Laplace transform rather than a finite Laplace series implicitly assumes the validity of such definition. The precise effect of this assumption remains to be deduced although, fortunately, because of the inverse relationship between β and ϵ , any such effect should be most pronounced at infinitely high energy.

where obviously $\epsilon_{\infty j} \simeq \epsilon_{\infty 1}$ and $0 < A_{\infty j} \ll A_{\infty 1}$ for all $j \geq 2$, and of course, $\epsilon_{\infty 1} \equiv \epsilon_\infty$, $A_{\infty 1} \equiv A_\infty$. We then have, using [32],

$$[38] \quad P(\epsilon)k(\epsilon) = \sum_j \mathcal{L}^{-1}\{A_{\infty j} \exp(-\beta\epsilon_{\infty j})\} * P|_\epsilon$$

$$[39] \quad = \sum_j A_{\infty j} P(\epsilon - \epsilon_{\infty j})$$

$$[40] \quad \simeq A_\infty P(\epsilon - \epsilon_\infty)$$

Thus, despite existing doubts, [37]–[40] demonstrate conclusively that the inverse transform of the experimental rate law is a good *approximation* to $P(\epsilon)k(\epsilon)$ in [33]. Equation 33 is not new, in the sense that it was first derived in the continuous form

$$[41] \quad k(\epsilon) = A_\infty \rho(\epsilon - \epsilon_\infty)/\rho(\epsilon)$$

by Slater (21). Yet, the obvious conceptual difficulty, that of the density of states at zero energy, is not encountered in the present derivation of [33]. The two $k(\epsilon)$ functions, [33] and [36], are quite different in form, as is shown for the example of cyclopropane in Fig. 1; however, they give rise to almost imperceptibly different rate laws given the present precision with which most thermal rate constants can be measured. In the present reformulation, we will adopt [33] for $k(\epsilon)$, based on two arguments presented in Appendix 1: briefly, the use of [33] always gives a better representation of the shape of the fall-off curve than does the use of [36], as is shown, for example, for cyclopropane in Fig. 2, and there is no other overriding reason for preferring [36] over [33].

VI. The Choice of the a_{ij}

The evaluation of the detailed transition probability matrix elements is clearly a formidable task. Yet, it is now well recognised that the macroscopic behaviour of a relaxation system depends predominantly on the overall relaxation rate and much less so on the pattern of the transition probabilities; this is especially so when the internal relaxation rate is many orders of magnitude faster than the reaction rate, as is the case with most thermal unimolecular reactions. Given this degree of freedom, it is desirable that our choice of the a_{ij} shall satisfy two criteria, one, that it leads to an algebraically tractable solution and two, that its physical implications are clear, so that the limits of its usefulness can be well understood. Figure 3a shows in schematic form a set of unperturbed eigenvalues for the internal relaxation of a molecule as a function of $[M]$, and Fig. 3b shows schematically the result of allowing this eigenvalue spectrum to be perturbed by a set of k_i elements. This is essentially the result obtained earlier in our numerical experiments (14) and foreseen a long

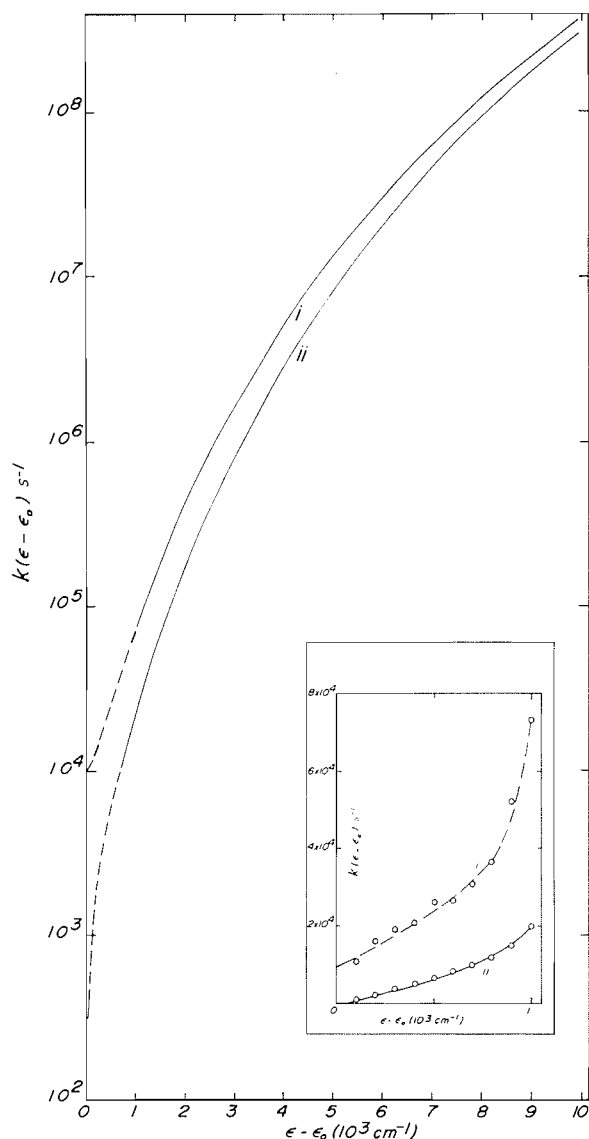


FIG. 1. $k(\epsilon)$ vs. ϵ for the isomerisation of cyclopropane above the critical energy. (i) Equation 33 with $A_{\infty} = 1.88 \times 10^{15} \text{ s}^{-1}$, $\epsilon_{\infty} = 22944 \text{ cm}^{-1}$, $\epsilon_{\text{act}} = \epsilon_{\infty}$ at all T . (ii) Equation 36 with $B' = 1.33 \times 10^{12} \text{ s}^{-1} \text{ cm}^{-1}$, $\epsilon_{\infty}^* = 22414 \text{ cm}^{-1}$; $\epsilon_{\text{act}} = \epsilon_{\infty}^* + kT$ and ϵ_{∞}^* is chosen such that $\epsilon_{\text{act}} = \epsilon_{\infty}$ at $T = 763.15 \text{ K}$. Both functions are step-like (on the scale of the figure) near the critical energy; the dashed segments represent smoothed functions. The inset shows the respective curves near threshold in linear scale.

time ago by Buff and Wilson (40): the important point for our argument is the wide separation between the bundle of eigenvalues representing the internal relaxations and the one representing the reaction rate. We propose, principally on the grounds of algebraic tractability, but also because no great injustice is done to the physical reality (as we shall argue below), to assume that the internal relaxations

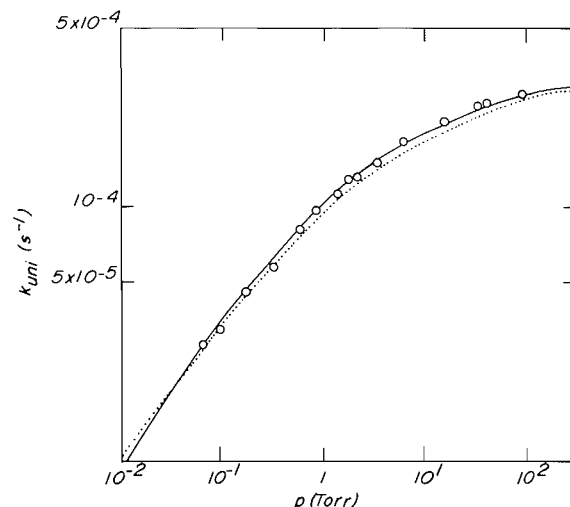


FIG. 2. Fall-off curves for the isomerisation of cyclopropane at 490°C . $\circ$ Experimental data of Pritchard, Sowden, and Trotman-Dickenson (57, 58); — theory using strict Arrhenius rate law, eqs. 33 and 34; theory using modified Arrhenius rate law, eqs. 35 and 36.

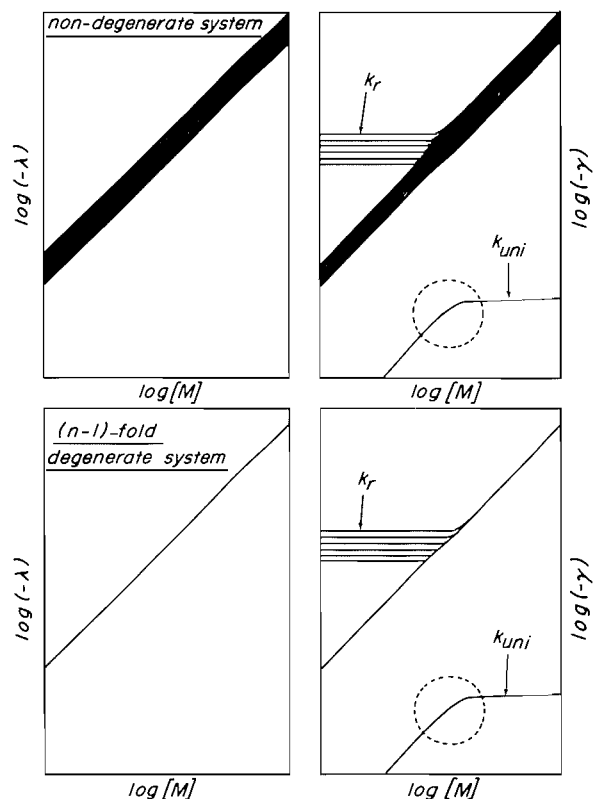


FIG. 3. Schematic representation of the eigenvalue behaviour of (a) a general relaxation matrix, (b) a general reaction matrix, (c) a relaxation matrix with $(n-1)$ -fold degeneracy (eq. 42), and (d) a reaction matrix with relaxation behaviour of (c). Circles indicate pressure region of interest in unimolecular reaction studies.

can be treated as being completely degenerate, Fig. 3c. The set of transition probabilities corresponding to this $(n-1)$ -fold degeneracy of the eigenvalues is (16, 41)

$$[42] \quad a_{ij} = \bar{\lambda}Z[M]\{(1 - \delta_{ij})\tilde{n}_j - \delta_{ij}(1 - \tilde{n}_i)\}$$

where $0 < \bar{\lambda} \leq 1$; for such a system, all the non-zero eigenvalues of the symmetrised transition matrix $\mathbf{B}$ will be degenerate, i.e.

$$\lambda_{n-2} = \lambda_{n-3} = \dots = \lambda_0 = -\bar{\lambda}Z[M] = -\mu < \lambda_{n-1} = 0$$

The quantity $\bar{\lambda}$ is therefore a (unknown) collisional efficiency parameter, analogous to the collisional deactivation efficiency (usually denoted by λ) in conventional theory (5, 6): it will be used here in a similar fashion, to effect a lateral shift in the calculated fall-off curve so as to obtain the optimum coincidence with the experimental observations; the quantity μ is of course the time constant for the internal relaxation.

It is important to understand clearly the implications of this approximation. Using [42], the unperturbed system relaxes in a purely exponential manner (16, 41), and it is this property which gives rise to a drastic simplification in the subsequent algebra. When the series of k_i elements is introduced, the behaviour of the perturbed eigenvalues changes to that depicted schematically in Fig. 3d, and the contention is that the encircled portion of the curve for k_{uni} in this diagram will be a good approximation to the encircled portion of the curve for k_{uni} in Fig. 3b; this is because the bundle of internal-relaxation eigenvalues is so well separated from the reaction-rate eigenvalue, and of course, as far as thermal unimolecular reactions are concerned, we are not interested in the eigenvalue behaviour elsewhere on these diagrams. In physical terms, the set of probabilities given by [42] will lead to a population evolution in which there are no fluctuations during the transient phase (42), but since we are interested only in the steady state in considering thermal unimolecular reactions, this artefact is unlikely to have serious consequences. On the other hand, examination of [42] reveals that the probability of an inelastic transition from state i to state j is proportional to the equilibrium population of the final state: this means that [42] will tend to overestimate the probabilities of transitions to states of high j , and we will return to a discussion of this problem later in the paper.

VII. The Unimolecular Rate Constant, k_{uni}

We are now interested in the way in which the presence of predissociation terms perturbs the

solutions of [42]. In the absence of any perturbation, the behaviour of these solutions is very simple (16), namely that both the total energy $\xi(t)$ and each of the individual populations $n_i(t)$ evolve in a pure-exponential way, i.e.

$$[43] \quad \frac{d\xi(t)}{dt} = \mu[\xi(t) - \bar{\xi}]$$

$$[44] \quad n_i(t) = \tilde{n}_i + [n_i(0) - \tilde{n}_i]e^{-\mu t}$$

Making use of the orthogonality relations between the vectors, together with eq. 15, it then follows that if we have r decay channels present, eq. 22 for the perturbed eigenvector elements simplifies to

$$[45] \quad \psi_{k,n-1} = \tilde{n}_k^{1/2} \gamma_{n-1}^{-1} \sum_l \tilde{n}_l^{1/2} d_l \psi_{l,n-1} + \mu^{-1} \sum_l (\delta_{kl} - \tilde{n}_k^{1/2} \tilde{n}_l^{1/2}) d_l \psi_{l,n-1}$$

where the index l runs over only the predissociating states. Extracting k_{uni} (i.e. $-\gamma_{n-1}$) out of this set of r equations, we have

$$[46] \quad k_{\text{uni}} = \frac{\sum_{i=1}^r a_i [\mu M]^{1-i}}{1 + \sum_{i=1}^r b_i [\mu M]^{-i}}$$

with

$$[47a] \quad a_i = \sum_{\{r\}} [(\sum \tilde{n}_j) \cdot (\prod k_j)]$$

$$[47b] \quad b_i = \sum_{\{r\}} [(1 - \sum \tilde{n}_j) \cdot (\prod k_j)]$$

where $\sum_{\{r\}}$ refers to summation over the $\{r\}$ combinatorial terms and the summations and products are over the combinatorial elements.² Equation 46 can clearly be regarded as a generalised Lindemann equation, but not the conventional generalised form [27].

In a real system, where there are millions of either discrete or quasi-continuous decay channels, eqs. 46 and 47 would become hopelessly cumbersome. It is therefore necessary to group the decay channels into

²The combinatorial terms appearing in eqs. 47a and 47b are defined as follows, for example

$$\begin{aligned} a_2 &= \sum_{\{2\}} \left[\left(\sum_j \tilde{n}_j \right) \cdot \left(\prod_j k_j \right) \right] \\ &= k_1 k_2 (\tilde{n}_1 + \tilde{n}_2) + k_1 k_3 (\tilde{n}_1 + \tilde{n}_3) + \dots \\ &\quad + k_1 k_r (\tilde{n}_1 + \tilde{n}_r) + k_2 k_3 (\tilde{n}_2 + \tilde{n}_3) + \dots \\ &\quad + k_2 k_r (\tilde{n}_2 + \tilde{n}_r) + \dots + k_{r-1} k_r (\tilde{n}_{r-1} + \tilde{n}_r) \end{aligned}$$

pseudo-levels, much in the same way as has been done previously in earlier master-equation formulations (5, 6, 14). However, in previous formulations, the graining procedure has been considered to apply to all states of the molecule, both reactive and unreactive, whereas here, we only need to coarse-grain the reactive levels into pseudo-levels: this is because, *in this approximation*, the information relating to the internal relaxation of the unreactive states has been incorporated into [46] through the use of [45]. This graining procedure is depicted schematically in Fig. 4, which is drawn in such a way as to emphasise the fact that the grain sizes need not correspond to equal energy gaps; the grain widths can be chosen in any way which optimises the calculation with respect to truncation errors and computing costs. In practice, the choice of the number of grains required and the grain widths is straightforward, and is based on the following observations. The first is that for all the reactions considered, 99% of the total rate comes from states within 30 kcal mol⁻¹ of the threshold, whatever the pressure, so that the summations in [46] can be terminated at about this energy. The second goes back to our earlier observation that if only one decay channel is present, the calculated fall-off curve is Lindemann in form, but as more decay channels are included in the calculation, the fall-off curve becomes more gradual (corresponding to a higher value of the Kassel shape parameter s) (14). This behaviour is shown in Fig. 5 for cyclopropane at 490°C with

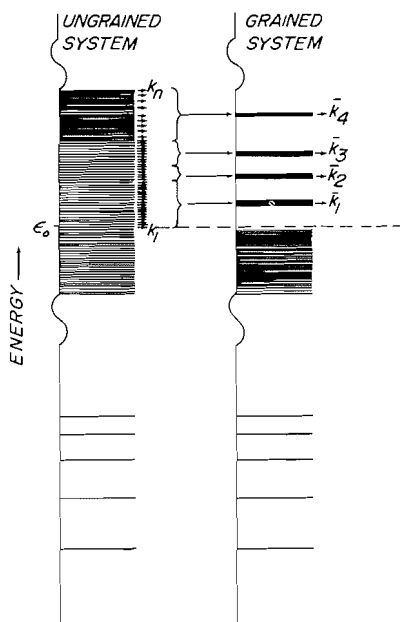


FIG. 4. Schematic representation of the graining procedure. Only the reactive states are grouped. The grain-widths need not be equal in energy.

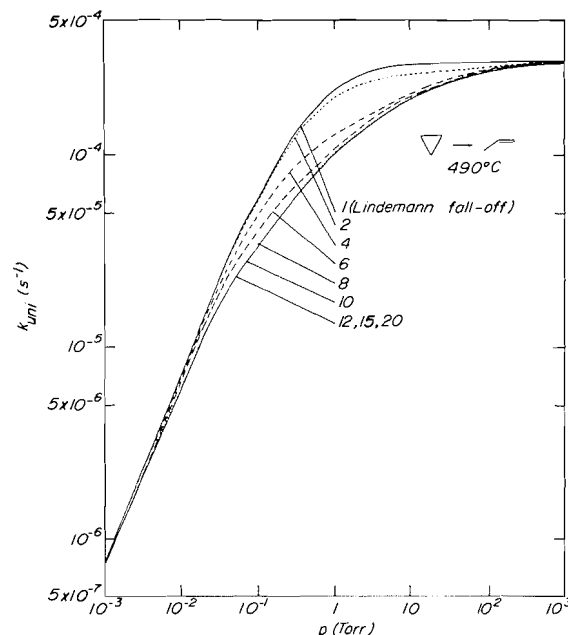


FIG. 5. Calculated fall-off curves for the isomerisation of cyclopropane at 490°C with $r = 1, 2, 4, 6, 8, 10, 12, 15$, and 20 pseudolevels. The last four curves are graphically indistinguishable. The first ($r = 1$) curve is exactly Lindemann in form. $A_\infty = 1.88 \times 10^{15} \text{ s}^{-1}$, $\epsilon_\infty = 65.6 \text{ kcal/mol}$.

$r = 1, 2, 4, 6, 8, 10, 12, 15$, and 20 decay channels: the calculated fall-off behaviour rapidly 'converges' as r increases, and in no case have we found it necessary to use more than 15 pseudo-levels to represent the spectrum of reactive states.

The use of this grouping procedure necessitates some redefinition of the meanings of terms used in eqs. 46 and 47 where the summations and products are over the individual predissociating states. For the ungrained system, the low-pressure and high-pressure limiting forms of the rate constant are

$$[48] \quad \lim_{[M] \rightarrow 0} k_{\text{uni}}(M) \equiv k_{\text{bim}}[M] = \mu \sum_r \bar{n}_r / (1 - \sum_r \bar{n}_r)$$

$$[49] \quad \lim_{[M] \rightarrow \infty} k_{\text{uni}}(M) \equiv k_\infty = \sum_r \bar{n}_r k_r$$

If we let $\bar{n}_r$ and $\bar{k}_r$ be the *grouped* equilibrium population and decay constant respectively for the *grained* system, we require that

$$[50] \quad \lim_{[M] \rightarrow 0} k_{\text{uni}}(M) = \mu \sum_j \bar{n}_j / (1 - \sum_j \bar{n}_j)$$

$$[51] \quad \lim_{[M] \rightarrow \infty} k_{\text{uni}}(M) = \sum_j \bar{n}_j \bar{k}_j$$

where the summations are now over the *grained* 'pseudo-levels'. Equations 48–51 may be satisfied if

we have, for each grain

$$[52] \quad \bar{n}_j = \sum_l \tilde{n}_l$$

$$[53] \quad \bar{k}_j = \sum_l \tilde{n}_l k_l / \bar{n}_j$$

where the summations l in [52] and [53] are only over the levels within each grain. Explicitly,

$$[54] \quad \bar{n}_j = [\mathcal{Q}(\beta)]^{-1} \sum_{\epsilon_l = \epsilon_j}^{\epsilon_{j+1}} g_l \exp(-\beta \epsilon_l)$$

$$[55] \quad \bar{k}_j = A_\infty [\mathcal{Q}(\beta) \bar{n}_j]^{-1} \exp(-\beta \epsilon_\infty) \times \sum_{\epsilon_l = \epsilon_j - \epsilon_\infty}^{\epsilon_{j+1} - \epsilon_\infty} g_l \exp(-\beta \epsilon_l)$$

However, as we will discuss further below, the mean width $\Delta \epsilon$ of the predissociating levels (arising from the uncertainty principle) is several orders of magnitude greater than the mean spacing between the levels for the molecules considered here (i.e. $\rho(\epsilon) \cdot \Delta \epsilon \gg 1$), so that [54] may readily be written in integral form, viz.

$$[56] \quad \bar{n}_j = [\mathcal{Q}(\beta)]^{-1} \int_{\epsilon_j}^{\epsilon_{j+1}} \rho(\epsilon) \exp(-\beta \epsilon) d\epsilon$$

where we define

$$[57] \quad \rho(\epsilon) = \sum_i P(\epsilon_i) \delta(\epsilon - \epsilon_i)$$

$\delta(\epsilon)$ being the Dirac delta function and the summation goes over all states i . Similarly [55] may be written as

$$[58] \quad \bar{k}_j = A_\infty [\mathcal{Q}(\beta) \bar{n}_j]^{-1} \exp(-\beta \epsilon_\infty) \times \int_{\epsilon_j - \epsilon_\infty}^{\epsilon_{j+1} - \epsilon_\infty} \rho(\epsilon) \exp(-\beta \epsilon) d\epsilon$$

Thus, the final expression for k_{uni} remains as eq. 46, but the expressions [56] and [58] for $\bar{n}_j$ and $\bar{k}_j$ are used in [47a] and [47b] instead of $\tilde{n}_j$ and k_j , and the index r is reinterpreted to be the number of pseudo-channels. For the remainder of this paper, for the reasons just mentioned above, the distinction between the discrete form for $k(\epsilon)$, eq. 33, and the continuous form, eq. 41, will be relaxed, and [33] and [41] will be used interchangeably.

A major advantage of this computational scheme is that once the grain size and the number of grains (usually 15) have been chosen, the unimolecular rate constant k_{uni} at any pressure can be evaluated explicitly, and the calculation needs not be started anew, as is the case with RRKM (and RRK) procedures, for each pressure: this is one of the two reasons why our method is so much more economical in computer time than the pre-existing procedures; the other reason is that our new procedure for cal-

culating densities and sums of states (43) can be up to two orders of magnitude faster than existing methods for complex molecules.

VIII. Which States to Count?

Our reformulation of unimolecular reaction theory is now complete, in a formal sense, the relevant equations being [46], [47a], [47b], [56], and [58]; however, the latter two equations contain the state-degeneracy of the molecule in the non-reactive region, and the density of states of the molecule in the reactive region. Tracing back the origins of this involvement of $P(\epsilon)$ and $\rho(\epsilon)$, it arises in our approach as part of a semi-empirical estimate (eq. 33) for the predissociation rate constants of the reactive states of the molecule, and our subsequent approximation using $\rho(\epsilon)$ for $P(\epsilon)$ in [57] which reduces [33] to the continuous form [41]. We have yet to consider in this context the problem of counting states, and to decide, for example, whether rotational or vibrational states can be regarded as making equal contributions to $k(\epsilon)$. Consequently, we have to examine the interrelationship between the state density and the fall-off properties for a molecule.

It has been recognised since the early days of unimolecular reaction theory that both the position and curvature of the fall-off curve are related to the complexity of the molecule. Usually, the number of atoms in the molecule, or the number of vibrational degrees of freedom, is used as an indicator of this complexity, but in fact we are really dealing with a manifestation of the *distribution* of state density in the molecule; a more useful measure of the complexity would be the number of vibrational degrees of freedom weighted by the reciprocals of their respective fundamental frequencies. The more complex a molecule is, the more vibrational degrees of freedom (of lower average frequency) it has, and hence the more drastic its increase in state density with energy. This causes the fall-off in unimolecular reaction rate with decreasing pressure to be more gentle in two ways. First, there are relatively more reactive states accessible by collisional or intramolecular energy transfer, so that at low pressure there will be a relatively higher rate of reaction, and hence a less drastic fall-off. Second, the more complex the molecule, the smaller is the ratio $P(\epsilon - \epsilon_\infty)/P(\epsilon)$ in eq. 33; thus, for fixed values of A_∞ and ϵ_∞ , $k(\epsilon)$ is smaller and the depletion of population by reaction at any particular pressure is not as severe.

It is of interest to note earlier attempts to characterise the fall-off behaviour of a unimolecular reaction in terms of the complexity of the reactant molecule, along such lines. These include the semi-empirical interpretation of the effective oscillator num-

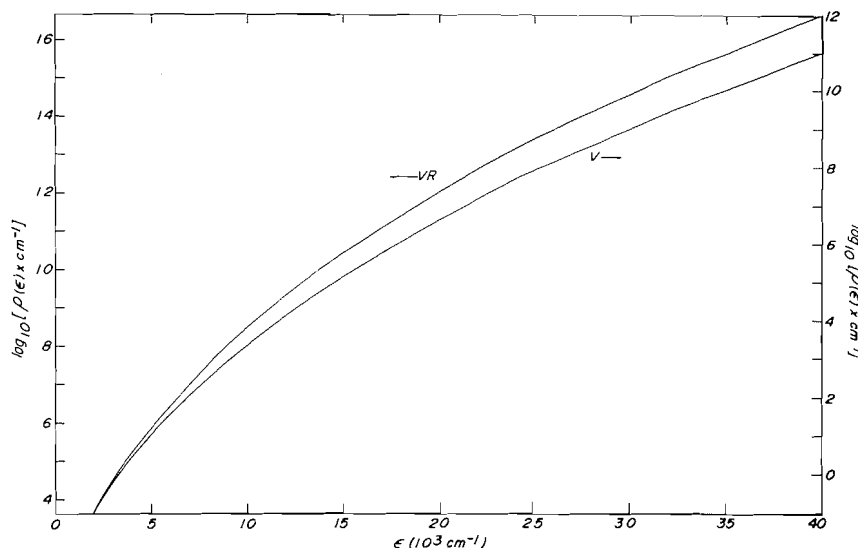


FIG. 6. Vibrational (V) and vibration-rotational (VR) state density of cyclopropane, using spectroscopic data in Appendix 3.

ber s in the Kassel integral in terms of the specific heat of the molecule (44), which is of course related to the distribution of energy levels (and the state density) in the molecule (43). The effective oscillator number s is no more than a fitting parameter (44, 45) whose role is essentially to calibrate the state density (albeit a classical and an inadequate one) of the reactant molecule. Yet the fact that most reactions consistently have a value of s equal to about half of the number of vibrational degrees of freedom tends to confirm the central role of state density in determining the fall-off behaviour, without the need to postulate additional parameters.

In conventional unimolecular rate theories, it has been customary to model a molecule as a collection of independent harmonic oscillators despite two longstanding difficulties. The first one is conceptual: a molecule consisting of pure harmonic oscillators can undergo neither dissociation nor internal energy randomisation. The second one is practical; until recently, fast, reliable methods for calculating state sums and state densities have been restricted to harmonic oscillators (46–48) and the apparent agreement between experiment and theory has certainly led to general acceptance of the harmonic oscillator model over the years, although this agreement has puzzled some authors (49). The actual distribution of states in a molecule is still very much unknown, especially at high energy. However, with the recent development of very efficient computational procedures for the calculation of state densities and state sums, both for harmonic and anharmonic oscillators (43, 50, 51), it is now possible to examine the conse-

quences of assuming such models, with and without rotation, and for a wide range of reactions.

Certainly rotational effects are important in diatomic dissociation (20), and rotational predissociation of polyatomic molecules, although poorly documented, is believed to be a common process (52). Indeed, there are no arbitrary reasons to exclude rotation, in the sense that in the transition probability matrix [42], the a_{ij} elements can equally well be regarded to exclude or include rotational transitions, either alone or simultaneously with vibrational transfer. The density of states for cyclopropane in the interesting energy range is shown in Fig. 6, both with and without rotation. It can be seen that the inclusion of rotational degrees of freedom increases the state density at high energy, and also changes the shape of the state-density curve: this will lead to a shallower fall-off curve, and a shift in the position of the fall-off to lower pressure for a fixed value of $\bar{\lambda}$.³

The consequence of inclusion of anharmonicity in the model reactant molecule is less obvious. Yet, it is clear that a minimal description of any unimolecular reaction should treat the molecule, at the very least, as a collection of independent *anharmonic* oscillators. This certainly would be more realistic than the usual harmonic oscillator approximation,

³The fall-off behaviour is, however, insensitive to the rotational constants of the molecule: indeed, in the present treatment, using the classical approximation for the rotational partition function and the steepest-descent method (53), the rotational constants only appear as multiplicative factors in $p(\epsilon)$ and $G(\epsilon)$, and they drop out of the final expression for k_{uni} .

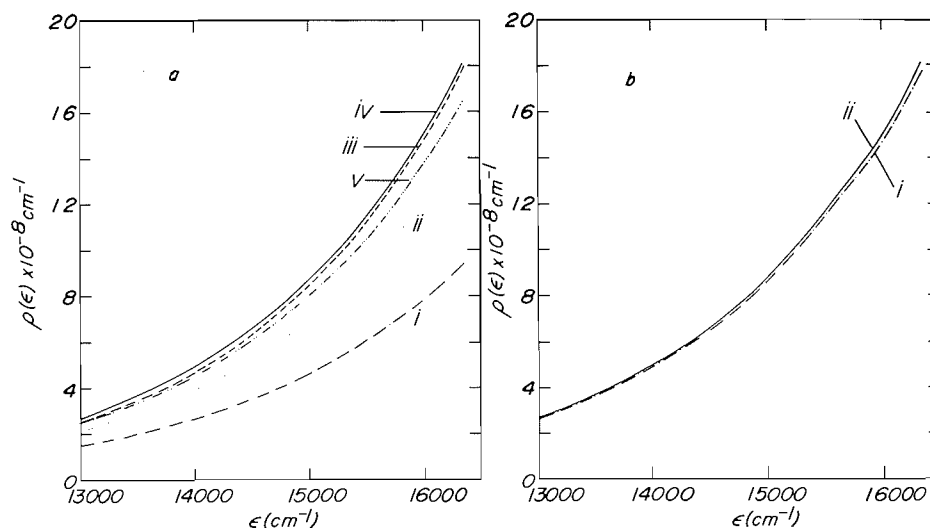


FIG. 7. (a) State density of methyl isocyanide in the reactive energy range, with the molecule modelled as (i) a set of harmonic oscillators and (ii)–(v) a set of independent Morse oscillators with $v_8 = 263 \text{ cm}^{-1}$ and $x_8 = 0, 0.01, 0.02$, and 0.03 respectively. All frequencies and anharmonicity factors are given in Appendix 3. All curves are essentially the same shape. (b) Variation of state density of methyl isocyanide in the reactive energy range with the anharmonicity factor of high-frequency vibrational mode $v_1 = 2966 \text{ cm}^{-1}$; $x_1 = 0$ (i) and 0.08 (ii) respectively. All other frequencies and anharmonicity factors are given in Appendix 3. Curves for $0 < x_1 < 0.08$ fall in between (i) and (ii).

and would remove the conceptual difficulties noted above: on the other hand, the longstanding success of harmonic oscillator models would appear to suggest that the additional complexity might be unnecessary, either because the fall-off behaviour is too insensitive to the anharmonicity, or else because of some fortuitous cancellation of errors. Moreover, anharmonicity data are very limited (54) indeed for the complex molecules of interest here, and so it is necessary to undertake a thorough examination of the effects of anharmonicity before we can have any confidence in the theoretical predictions. The information on anharmonicities that is available comes from first-overtone measurements and/or approximate theoretical calculations (55), and it would appear that with few exceptions, regarding the vibrations as Morse oscillators of frequency v_i , anharmonicity factors x_i lie in the range of 1–3%, i.e. $0.01 \leq x_i \leq 0.03$ (see Appendix 3 for definitions of v_i and x_i). Qualitatively, the effect of introducing this anharmonicity is to increase the state density in the intermediate energy range, but to reduce it at high energies, since the set of energy levels now converges to a maximum value of the vibrational quantum number $v_{i,\text{max}}$. Given the typical values of x_i , $v_{i,\text{max}}$ will commonly be between 15 and 50. In order to focus our attention on the nature of the problem, we may subdivide the vibration frequencies of molecules roughly into three classes, those around 300, 1000, and 3000 cm^{-1} . We note that the interesting energy range for thermal unimolecular reac-

tions is between 10 000 and 25 000 cm^{-1} : for example, the Arrhenius activation energy for the isomerisation of methyl isocyanide is about 13 360 cm^{-1} ; states in this energy range could either have only very few high-frequency quanta, or at the other extreme, some tens of low-frequency quanta. It follows therefore that the calculated state density $\rho(\epsilon)$ will not be very sensitive to variations in the anharmonicity factors for the high-frequency modes, but might be so for the low-frequency modes. No systematic study of this point has been reported in the literature, to our knowledge. We have therefore presented in Appendix 2 (and Figs. 7 and 8) such an analysis for methyl isocyanide and cyclopropane.

With the results displayed in Figs. 6–8, we can summarise the conclusions of this section as follows. The complexity of a reactant molecule can be identified with the distribution of energy states of the molecule, particularly at high energy, and this complexity property determines the shape of the fall-off curve, and also, more-or-less, its position. Through the operation of eqs. 46, 47, 56, and 58, the shape of the state-density function determines the shape of the fall-off curve, and the magnitude of the state density in the interesting energy range determines the position of the fall-off curve on the pressure axis, to within a factor of $\bar{\lambda}$. Clearly then, the position and shape of the fall-off curve are related, although not explicitly. Also, the calculated form of k_{uni} will depend upon whether $\rho(\epsilon)$ is determined assuming harmonic or anharmonic oscillators, with or without rotation: as

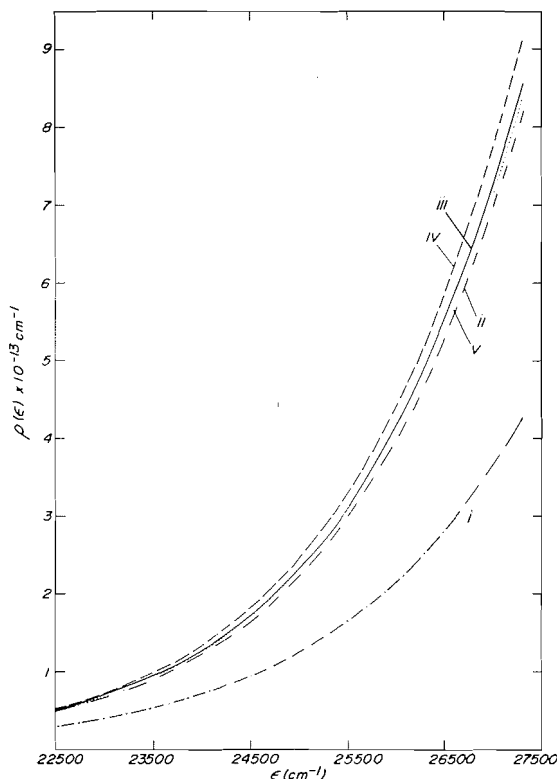


FIG. 8. Variation of state density of cyclopropane in the reactive energy range with the anharmonicity factor of the rocking mode ($\nu_8 = 854 \text{ cm}^{-1}$). (i) Harmonic-oscillator model (all modes); (ii)–(v) Morse-oscillator model (see Appendix 3), with $x_8 = 0, 0.015, 0.03$, and 0.09 respectively.

we shall find in the following sections, the best agreement between theory and experiment always results from the anharmonic oscillator model, with rotation. Fortunately, the sensitivity of the results to small imperfections in the assumed model is not great. Consequently, if the anharmonicity factors x_i can be estimated to within ± 0.01 , the shape of the $\rho(\epsilon)$ function will be of acceptable accuracy, and the absolute magnitude of the $\rho(\epsilon)$ function in the interesting energy range (which determines the pressure shift $\bar{\lambda}$) will be correct to within about 30–50%. Similarly, imperfections in calculating the rotational contribution to $\rho(\epsilon)$ will also be relatively unimportant: the model system of independent rotors and oscillators is an acceptable approximation.

IX. Comparison with Experiment

In most cases, experimental tests of the theory of unimolecular reactions have been confined to a comparison of the shapes of the experimental and theoretical fall-off curves; however, for some reactions, additional important information is available, namely the variation of the Arrhenius temperature coefficient

and/or of the kinetic isotope effect with pressure. In this section we will show that our reformulation gives consistently better agreement with experiment than do any existing theories of unimolecular reactions for five important thermal unimolecular reactions. In fact, a thorough test of the theory is only possible for the isomerisation of cyclopropane, because this is the only molecule for which fairly complete anharmonicity data are available, and it is from the detailed examination of this reaction that we came to the conclusions of the preceding section: that our state count must refer to anharmonic oscillators and include all rotational degrees of freedom. Guided by the anharmonicity data for cyclopropane, we have deduced a set of rules for *estimating* anharmonicity factors for other molecules (see Appendix 3): with these estimated anharmonicity factors, our method gives virtually exact agreement with experiment for the thermal decompositions of cyclobutane and ethyl chloride, and it gives better agreement than exists at present for the isomerisations of methyl isocyanide and ethyl isocyanide (and in these two cases, we believe there may be valid reasons for the remaining discrepancies). The required molecular constants and other parameters are collected together in Appendix 3 and straightforward and efficient computer programs for the calculation of k_{uni} therefrom are described in Appendix 4. In this section, we will confine our attention to the description of the results of these calculations, and the comparison of the computed results with experiment; for the sake of brevity we will take as read earlier extensive comparisons of this nature, particularly those of Wieder and Marcus (3, 4), Lin and Laidler (56), and Robinson and Holbrook (5).

A. The Shape of the Fall-off Curve

Figure 9 shows the experimental results for the thermal isomerisation of cyclopropane at 490°C , obtained by Pritchard, Sowden, and Trotman-Dickenson (57, 58), compared with three model calculations for k_{uni} . The model calculations use the spectroscopic constants of Cyvin (55), the value of ϵ_∞ found by Falconer, Hunter, and Trotman-Dickenson (59), a value of k_∞ found by extrapolating k vs. $1/p$ (58), and A_∞ consistent with k_∞ and ϵ_∞ (see Appendix 3 for details). The dotted line shows a conventional RRKM calculation with these parameters (5), using harmonic oscillators and no rotation. The dashed curve shows the results of the present reformulation, again using harmonic oscillators with no rotation; the difference between these two curves stems, in part, from the difference between eq. 27 and eq. 46 and, in part, from the difference in the respective forms of the $k(\epsilon)$ function (cf. Fig. 2). Thirdly, the

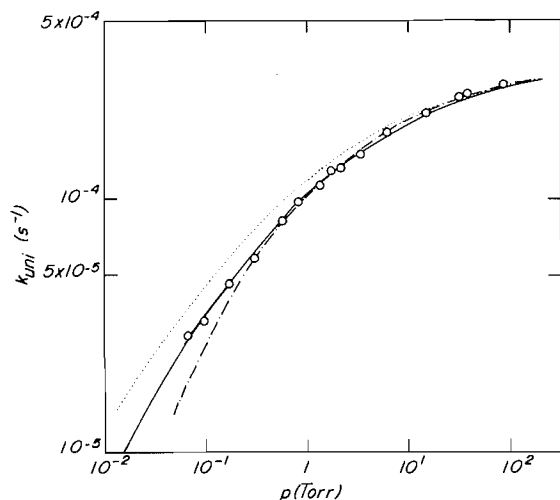


FIG. 9. Thermal isomerisation of cyclopropane at 490°C: ○ Experiment of Pritchard, Sowden, and Trotman-Dickenson (57, 58); ... RRKM theory, with the antisymmetric CH stretch ($\nu = 3103 \text{ cm}^{-1}$) mode isolated as reaction coordinate; ---- this formulation, harmonic oscillator; — this formulation, rotating Morse oscillator.

solid line shows the results of the present reformulation using Morse oscillators and including rotation: other alternative choices, viz. Morse oscillators without rotation or harmonic oscillators with rotation are not as good, and are omitted for the sake of clarity. We remind the reader at this point that all three theoretical curves in Fig. 9 have been subjected to a lateral (pressure) shift in order to bring them into the best coincidence with the experimental data: the magnitudes of these shifts will be discussed separately below.

Figure 10 shows the experimental results for the thermal decomposition of cyclobutane to two ethylene molecules at 448°C, again obtained by Pritchard, Sowden, and Trotman-Dickenson (58, 60); other experimental data are very closely coincident (5, 61, 62) and are omitted for clarity. The solid line represents the form of k_{uni} calculated from the present reformulation, suitably shifted, and using the molecular constants and reaction-rate parameters given in Appendix 3.

Another reaction which has attracted much attention is the thermal decomposition of ethyl chloride to form ethylene and hydrogen chloride (5); the data are, however, much more scattered than the cyclopropane or cyclobutane data, and not always concordant (5, 63). Figure 11 shows the experimental data of Holbrook and Marsh (64) at 521°C, compared with our calculated form of k_{uni} ; the agreement is equally satisfactory at other temperatures also, but unfortunately, because of the scatter inherent in the experimental measurements, does not provide a very stringent test of this reformulation.

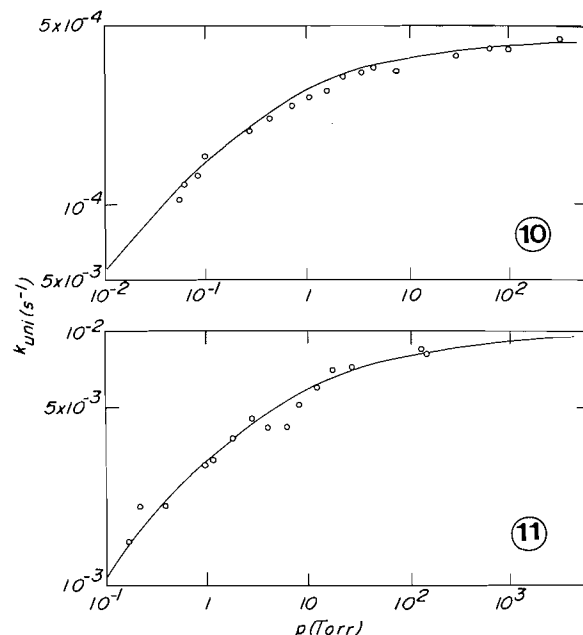


FIG. 10. Thermal decomposition of cyclobutane at 448.3°C: ○ Experiment of Pritchard, Sowden, and Trotman-Dickenson (58, 60); — calculation, eq. 33.

FIG. 11. Thermal decomposition of ethyl chloride at 521°C: ○ Experiment of Holbrook and Marsh (64); — calculation, eq. 33.

The thermal isomerisation of methyl isocyanide to methyl cyanide represents, to date, the most thoroughly studied unimolecular reaction (5, 9, 34). The fall-off behaviour was first studied by Schneider and Rabinovitch (9) at 200, 230, and 260°C; they also derived the infinite-pressure Arrhenius parameters as $A_{\infty} = 10^{13.6} \text{ s}^{-1}$ and $\epsilon_{\infty} = 38.4 \text{ kcal mol}^{-1}$. Unfortunately, the isomerisation is an exothermic reaction, and corrections had to be applied to the high-pressure data to allow for self-heating. Collister and Pritchard (34) have recently studied this reaction over the temperature range 120–320°C, in the rigorous absence of self-heating. The highest pressure they used in any of their experiments was 100 Torr and in the region of overlap of the two studies, i.e. $2 \leq p \leq 100 \text{ Torr}$, $200 \leq T \leq 260^{\circ}\text{C}$ the agreement is excellent. Collister and Pritchard extrapolated their rate data (using an iterative RRKM procedure to find the most consistent set of ϵ_{∞} and k_{∞}) to give $A_{\infty} = 10^{13.35 \pm 0.11} \text{ s}^{-1}$ and $\epsilon_{\infty} = 38.2 \pm 0.2 \text{ kcal mol}^{-1}$, apparently in excellent agreement also with the values given by Schneider and Rabinovitch; however, Schneider and Rabinovitch's values of k_{∞} are about 50% greater than those of Collister and Pritchard, which complicates the comparisons somewhat.

Schneider and Rabinovitch (9) examined a whole series of reaction models, but were never able to obtain consistent agreement between the theoretical and

experimental fall-off curves (cf. also ref. 5), nor were the pressure shifts monotonic (as they would be expected to be) at the three experimental temperatures. The shape of the fall-off at 230°C has been re-examined by Forst (23, 24) confirming that the experimental and theoretical shapes are not the same. Figure 12 shows the theoretical fall-off curve calculated by our new procedure, including anharmonicity and rotation, together with the experimental points of Schneider and Rabinovitch (9); again, the calculated and experimental curves do not have the same shape, and in the way we have adjusted the pressure shift, there seems to be a sudden rise in the rate constant above 300 Torr. We believe there could be a logical explanation of this observation. Schneider and Rabinovitch estimated the degree of self-heating in their high-pressure reactions on the basis of a ΔH of isomerisation of $-15 \text{ kcal mol}^{-1}$ and a thermal conductivity of $0.25 \text{ mW cm}^{-1} \text{ deg}^{-1}$. We have recently remeasured both ΔH as $-23.70 \pm 0.14 \text{ kcal mol}^{-1}$ (65) and the thermal conductivity, about $0.20 \text{ mW cm}^{-1} \text{ deg}^{-1}$ at 230°C (66), which implies that all of the correction factors for self-heating estimated by Schneider and Rabinovitch are *too small by about a factor of two*. It seems likely that if sufficient raw experimental data are still available to enable these self-heating corrections to be calculated properly, some of the present discrepancies might be eliminated. As will be shown below, our calculated curve for k_{uni} fits the data of Collister and Pritchard (34)

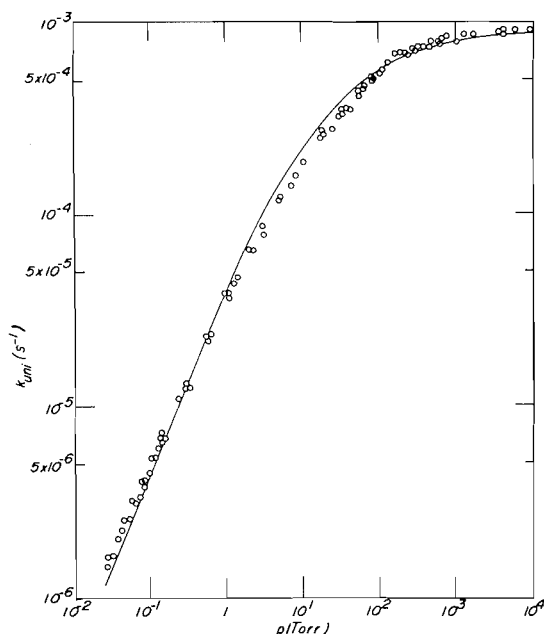


FIG. 12. Thermal isomerisation of methyl isocyanide at 230.4°C: ○ Experiment of Schneider and Rabinovitch (9); — calculation, eq. 33.

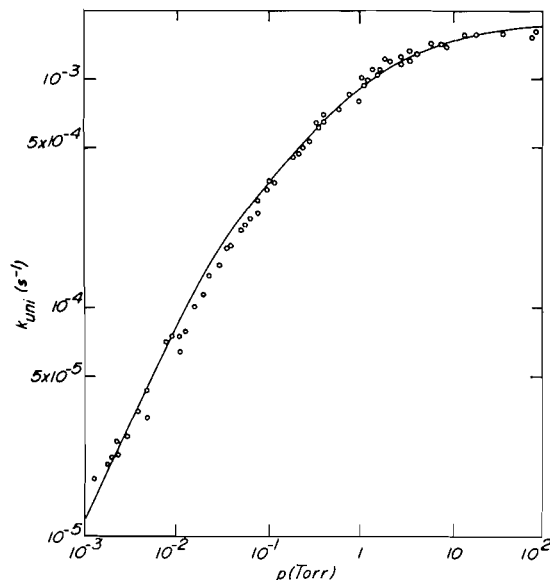


FIG. 13. Thermal isomerisation of ethyl isocyanide at 230.9°C: ○ Experiment of Maloney and Rabinovitch (10); — calculation, eq. 33.

very well, but in view of the short pressure range of this work ($2 \leq p \leq 100 \text{ Torr}$), this again is not a very stringent test of the new theory.

Figure 13 shows the experimental data of Maloney and Rabinovitch (10) for the thermal isomerisation of ethyl isocyanide at 231°C, plotted together with the theoretical k_{uni} curve calculated from the present reformulation. There is a similar discrepancy to that shown for methyl isocyanide in Fig. 12, which again could be interpreted as an unexpected rise in the experimental rates above 1 Torr. As before, the self-heating corrections are suspect, and should be re-evaluated on the basis of the new value for ΔH (about $-22 \text{ kcal mol}^{-1}$) (65) and a new estimate for the thermal conductivity; however, self-heating should not be important at pressures as low as 1 Torr, so there may be other, as yet undiscerned, reasons for this poor agreement.

B. The Position of the Fall-off Curve

As we noted earlier, and in common with other theories of unimolecular reactions, the calculation of k_{uni} does not establish the absolute position of the fall-off curve on the pressure axis. In conventional theory, a collisional deactivation efficiency parameter λ is used to move the calculated fall-off curve along the pressure axis to obtain the best fit with the experimental data. However, this collision efficiency parameter has two disconcerting properties.

The first is that it either varies widely with temperature, or does not vary in a monotonic fashion with temperature, and thus it does not appear to have the kind of properties required of an energy-

transfer probability. For example, for 1,1-dichlorocyclopropane (5), a value of $\lambda = 1.0$ would appear to be satisfactory near 360°C, whereas $\lambda = 0.4$ fits the data better at 425°C: in view of the common occurrence of values of λ of the order of unity, this large variation over such a short temperature range seems surprising. Better documented examples come from experiments on the methyl isocyanide and ethyl isocyanide isomerisations where the pressure correction factors (with the 'E-300' model) (9, 10) at three consecutive temperatures are CH_3NC : 0.91, 1.11, and 0.95; $\text{C}_2\text{H}_5\text{NC}$: 0.88, 1.37, and 1.25 respectively. It seems unlikely that energy-transfer collision efficiencies would vary in this random fashion, and it is not entirely clear to us whether recalculation of the self-heating corrections (alluded to above) would remove these anomalies altogether. That they are in fact anomalies is shown in Fig. 14 where we plot the data of Collister and Pritchard (34) for three different temperatures: the three lines which pass through the experimental data points all have the same value of the shift parameter $\bar{\lambda}$, and they all fit equally well. The representation of these agreements in this fashion does, however, require some amplification. We do not know how the relaxation time constant μ (cf. eqs. 42–44) for the internal relaxations varies with temperature: in the absence of more definite information, the least biased guess we can make is that $\mu = cT^{1/2}$ (note that $\mu = \bar{\lambda}Z[M]$), and it is this assumption that brings all the experimental data and the theoretical lines into coincidence with a single value of $\bar{\lambda}$. Clearly, if μ were assumed to be constant, the $\bar{\lambda}$ would be monotonic with temperature, unlike the earlier data (9, 10): thus, it seems that this particular difficulty may be mainly experimental in origin.

The second disconcerting property of the collision deactivation efficiency parameter λ is that it often seems to have a value very close to unity. The precise significance of λ is of course unclear because its numerical magnitude must depend upon the choice of the collision rate. It is well recognised that collision rates must be internal-state dependent, and that the 'structureless' collision numbers in common use represent nothing more than macroscopically averaged quantities. Clearly this averaging is very insensitive to the detailed distribution of collision efficiencies, and in the absence of more specific information, we can usually do no more than to use normal gas-kinetic collision diameters in calculating these collision rates (67). Given these difficulties, the occasional occurrence of values of λ slightly in excess of unity, by perhaps 10 or 20% (57, 60) is not alarming, and their observation has actually been interpreted as being supportive of the concept of unit deactivation probability (68). However, in very-low pressure experi-

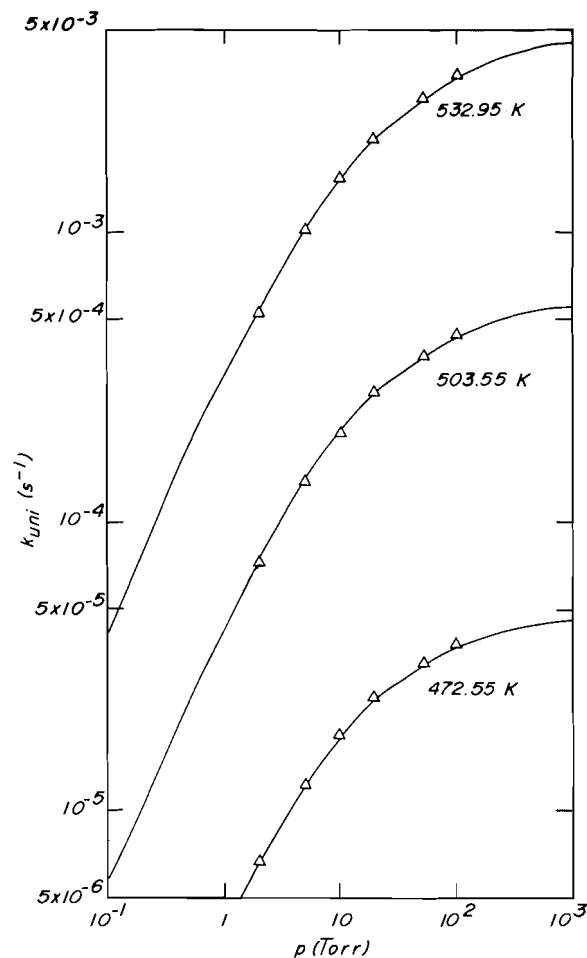


FIG. 14. Thermal isomerisation of methyl isocyanide at 472.55 K, 503.55 K, and 532.95 K. Δ Experiment of Collister and Pritchard (34, 35); — eq. 33.

ments, where activation takes place predominantly at the wall rather than in gaseous collisions, molecule-wall collisions appear to be some 5–10 times as efficient as molecule-molecule collisions (11–13). Some doubts have been voiced about one of these experiments, and it was alleged (69) that the levelling off of the rate constant at low pressure, originally ascribed to wall activation, was really due to water vapour outgassing to a constant pressure from the vessel walls; the critics omitted to observe that a trap at -80°C was permanently in contact with the reaction vessel during each low-pressure run, in order to suppress unwanted pressures of both mercury and water vapours. Let us assume that accommodation coefficients for internal energy of complex molecules at the surface are of the order of 1 in a few collisions: to our knowledge, the only measurements which exist (70) relate to simple molecules at room temperature, but one can easily build an argument based

TABLE 1. Calculated collision efficiency parameters $\bar{\lambda}$

Molecule	T (K)	m (g/mol)	σ (Å)	$10^{10} \times Z$ (cm ³ molecule ⁻¹ s ⁻¹)	$10^{11} \times \mu'$ (cm ³ molecule ⁻¹ s ⁻¹)	$\bar{\lambda}$
CH ₃ NC	472.6	41.0	4.5	4.4 ₄	4.31	0.09 ₇
	503.6			4.5 ₉	4.86	0.10 ₆
	533.0			4.7 ₂	5.26	0.11 ₁
C ₂ H ₅ NC	504.1	55.1	5.5	5.9 ₂	6.91	0.11 ₇
C ₃ H ₆	763.2	42.1	4.0	4.4	4.43	0.10
C ₄ H ₈	721.5	56.1	4.5	4.7	4.22	0.09 ₉
C ₂ H ₅ Cl	794.2	64.5	4.5	4.6	4.77	0.10 ₄

on the results of these measurements in favour of our assumption. Accepting that the low-pressure kinetic experiments (11–13) demonstrate molecule–wall collisions to be almost an order of magnitude more efficient than molecule–molecule collisions, then we might guess that λ would lie in the range $10^{-1.5}$ to 10^{-1} and that, certainly, λ would not be unity as it generally appears to be in RRKM treatments.

The use of the master-equation formulation enables us to shed some light on this problem, although in a cascading relaxation system the concept of a collision efficiency is not always a well-defined one, since the observed relaxation time depends upon the initial distribution (17). However, it takes on a theoretical meaning when, as in our reformulation, the internal relaxation can be described by a single relaxation time: in our case, cf. eq. 42, the ‘sum rule’ is obeyed (16) whence, unambiguously

$$[59] \quad \mu = \bar{\lambda}Z[M] \quad \text{or} \quad \bar{\lambda} = \mu/Z[M]$$

where Z , the collision number, is defined as in eq. 1, or since (cf. ref. 5, p. 164) M is usually another molecule of the same kind,

$$[60] \quad Z = 4\sigma^2(\pi kT/m)^{1/2}$$

where m is the mass of the molecule. It is also convenient to define a relaxation rate constant

$$[61] \quad \mu' = \mu/[M] = \bar{\lambda}Z$$

Given the fact that there is excellent coincidence between the experimental and theoretical fall-off curves for the five molecules considered in this paper, the determination of μ' is straightforward, and the results are shown in Table 1. It appears that all the collision efficiency parameters $\bar{\lambda}$ are of the order of 0.1, which is certainly more in line with modern ideas about energy transfer (11–13, 70) than is the unit efficiency of the RRKM theory.

This could, however, be an artefact (but we think not). The a_{ij} in eq. 42 overestimate the importance of large-energy transitions with respect to small-energy transitions, so that all cascading processes coalesce into a single process; however, experiments on these

molecules have not yet resolved the cascading processes, so clearly this contraction is reasonably acceptable. The effect of overemphasising large-energy transitions is clearly shown in Fig. 15 which shows the population depletion in the reactive region at low-pressure, i.e. $\bar{n}_i/\bar{n}_i$, so that the equilibrium distribution is represented by $\bar{n}_i/\bar{n}_i = 1$. It can be seen that the populations of states begin to fall away below the threshold ϵ_0 , and in a real system should continue to fall away to zero at very high energies (5, 14); in our case, however, somewhat above the topmost predissociating state, the values of $\bar{n}_i/\bar{n}_i$ return to the equilibrium value, because of the strength of the large-energy transitions implicit in using [42]. This presents no difficulty in our numerical formulation though, since we continue to include predissociating states up to a sufficiently high energy that there is no further contribution to the total rate, as shown in the shaded portion of the diagram. The only point we make here, then, is that the interpretation of the physical meaning of $\bar{\lambda}$ still requires a little circumspection.

C. The Pressure Dependence of the Activation Energy

The pressure dependence of the activation energy provides a sensitive test for unimolecular rate theories. At any pressure p and inverse temperature β , the formal expression for the activation energy

$$[62] \quad \epsilon_{\text{act}}(\beta; p) = -d \ln k_{\text{uni}}(\beta; p)/d\beta$$

may be obtained directly using the Hellman–Feynman theorem,

$$[63] \quad \epsilon_{\text{act}}(\beta; p) = (k_{\text{uni}})^{-1} \psi_{n-1}^T (\partial B / \partial \beta) \psi_{n-1}$$

The evaluation of [63] for a general system is clearly a formidable task, although some general properties regarding the pressure dependence of ϵ_{act} have been derived by various authors (6, 71, 72). It is simpler, in fact, to compute ϵ_{act} directly from the analytic expression for k_{uni} , if it is available. Indeed, in the present treatment, since ϵ_{act} is only weakly temperature-dependent at all pressures (and independent of temperature at infinite pressure by virtue of the strict

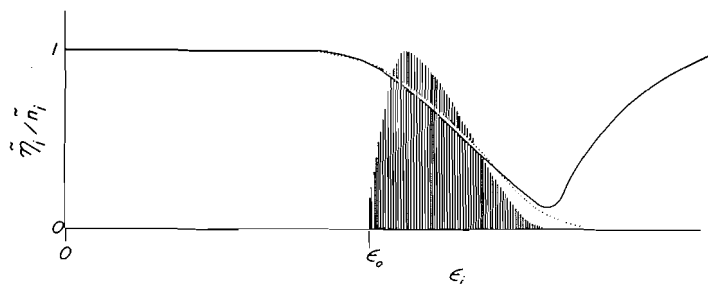


FIG. 15. Schematic representation of population depletion of states $\tilde{n}_i/\bar{n}_i$ at low pressure as a function of reactant energy ϵ_i , in the $(n-1)$ -fold degenerate system (solid line) and in a realistic system (dotted line). ϵ_0 is the critical energy. The shaded area is the distribution profile of reacting molecules.

Arrhenius behaviour), it may be computed from

$$[64] \quad \epsilon_{\text{act}}(\beta; p) = \frac{1}{\Delta\beta} \left\{ 1 - \frac{k_{\text{uni}}(\beta + \Delta\beta; p)}{k_{\text{uni}}(\beta; p)} \right\}$$

where $\Delta\beta$ is a small change in the inverse temperature β .

The most extensive data available on the pressure dependence of ϵ_{act} are for the thermal isomerisation of methyl isocyanide, from the work of Schneider and Rabinovitch (9) and of Collister and Pritchard (34); in the former work, the activation energy at low pressure was regarded as being temperature-independent, but in the latter work, ϵ_{act} was represented as being both pressure-dependent and temperature-dependent, as required by the present theoretical framework. Figure 16 compares the calculated values of ϵ_{act} from eq. 64 at 230°C with the experimental data of Collister and Pritchard (for that temperature) and of Schneider and Rabinovitch (mean values derived from Arrhenius plots over the range 200–260°C). The agreement between the present

theory and experiment is excellent, and is superior to the RRKM description.

Figure 17 shows a similar comparison, using the experimental data of Maloney and Rabinovitch (10), for ethyl isocyanide: the solid line is the present theory, and the two dashed lines are those given by two RRKM models calculated by Maloney and Rabinovitch. The agreement here is not as good as it was for methyl isocyanide, but this is to be expected because the pressure shifts are more irregular for ethyl isocyanide (see preceding subsection). The present theory also gives a very good representation of $\Delta\epsilon_{\text{act}} = \epsilon_{\text{act}}(\infty) - \epsilon_{\text{act}}(0)$: Maloney and Rabinovitch estimated this to be about 4 kcal mol⁻¹ compared with our calculated value of 4.5 kcal mol⁻¹.

Figure 18 compares theory and experiment for cyclobutane: the experimental data are deduced from the low-pressure rate constants of Vreeland and Swinehart (62) in the range 410–500°C and the high-pressure rate constants of Carr and Walters (73) between 419 and 460°C. The theoretical curve is calculated for 448°C, and the agreement is again excellent. However, the calculated value of $\Delta\epsilon_{\text{act}}$ cannot be compared satisfactorily with experiment because

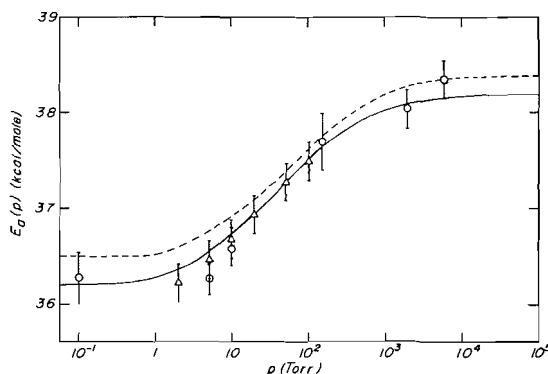


FIG. 16. Variation of activation energy ϵ_{act} with pressure of CH_3NC isomerisation at 230°C: $\circ$ Experiment of Schneider and Rabinovitch (9); $\triangle$ experiment of Collister and Pritchard (34); --- RRKM calculation (9) with $\epsilon_\infty = 38.4$ kcal/mol; — present formulation, eq. 64 with $\epsilon_\infty = 38.2$ kcal/mol; n.b. activation energy at low pressure is insensitive to the choice of ϵ_∞ .

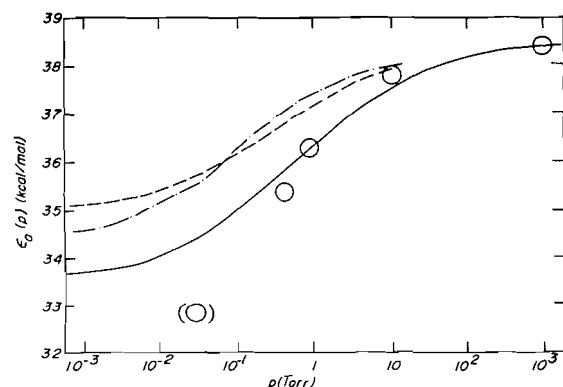


FIG. 17. Variation of activation energy ϵ_{act} with pressure of $\text{C}_2\text{H}_5\text{NC}$ isomerisation at 231°C: $\circ$ Experiment of Maloney and Rabinovitch (10); --- and --- RRKM calculations for two models (10); — present formulation, eq. 64.

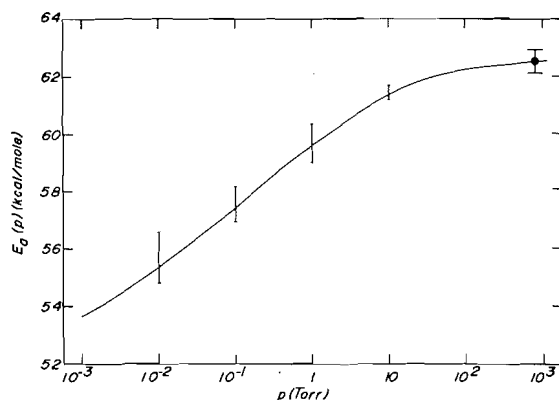


FIG. 18. Variation of activation energy ϵ_{act} with pressure for the decomposition of cyclobutane at 448.3°C. $\square$ value derived from experimental data of Vreeland and Swinehart (62) between 410 and 500°C; $\frac{1}{2}$ data of Carr and Walters (73); — present formulation, eq. 64.

of the abnormal levelling off of the rate data below 10^{-3} Torr.

Paradoxically, similar experiments have not been performed for cyclopropane: however, an approximation to $\Delta\epsilon_{\text{act}}$ is available from the difference between the high-pressure (59) and low-pressure (11) activation energies, giving $8.4 \pm 2 \text{ kcal mol}^{-1}$ in the range 477–517°C compared with the theoretical value of $\Delta\epsilon_{\text{act}} = 8.2 \text{ kcal mol}^{-1}$, at 490°C.

D. The Pressure Dependence of the Kinetic Isotope Effect

The kinetic isotope effect is an immediate consequence of any statistical rate theory based on molecular densities of states, such as the present formulation. The quantum-statistical effect at low pressure, from the RRKM point of view, has been described by Rabinovitch, Setser, and Schneider (74). Our reformulation, not unexpectedly, leads to a similar limiting form for the isotope ratio $\gamma = k_{\text{uni},a}/k_{\text{uni},b}$ (for isotopes a and b) at low pressure, except that the respective rates $k_{\text{uni},a}(p \rightarrow 0)$ and $k_{\text{uni},b}(p \rightarrow 0)$ are governed by the relaxation rate constants μ' rather than the collision numbers Z , whence

$$[65] \quad \gamma(p \rightarrow 0) = \frac{\mu'_a Z_b}{\mu'_b Z_a} \cdot \frac{\sum \bar{n}_{r,a} (1 - \sum \bar{n}_{r,b})}{\sum \bar{n}_{r,b} (1 - \sum \bar{n}_{r,a})}$$

Although Z_a and Z_b are simply related (via the square roots of the molecular masses), there is no need for μ'_a and μ'_b to be similarly related. However, since isotope-effect experiments are customarily performed by internal comparison, the required value of μ'_a would have to be that appropriate to the relaxation of species a by the mixture of isotopic molecules, and likewise for μ'_b ; under these conditions, it seems likely that μ'_a/μ'_b would be close to unity, and with

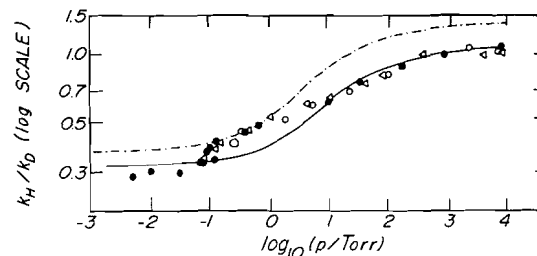


FIG. 19. Kinetic isotope ratio $k_{\text{H}}/k_{\text{D}}$ of CH_3NC , CD_3NC isomerisation vs. pressure p . The three symbols refer to three different groups of experiments of Schneider and Rabinovitch (75). — Present formulation, eq. 65; RRKM calculation using the 'E-300' model of Schneider and Rabinovitch (75).

this simplifying assumption, the calculation of γ at all pressures is straightforward if the infinite-pressure Arrhenius parameters are known.

The isomerisation of trideuteriomethyl isocyanide has been studied by Schneider and Rabinovitch (75) at 230°C and over a range from 10^{-2} Torr to greater than 10^4 Torr; the isotope ratio γ varied from 0.28 at the low pressures to 1.07 at the highest pressures. The solid line in Fig. 19 shows the ratio $\gamma(p) = k_{\text{H}}(p)/k_{\text{D}}(p)$ calculated using eq. 65 and using the observed values of $\gamma(\infty) = 1.07$ and $\epsilon_{\text{act,D}}(\infty) - \epsilon_{\text{act,H}}(\infty) = 0.1 \text{ kcal mol}^{-1}$. The dotted line is the 'E-300' RRKM model of Schneider and Rabinovitch (75) based on the same values of $\gamma(\infty)$ and $\Delta\epsilon_{\text{act}}(\infty)$. The agreement with experiment of the reformulation, eq. 65 is again superior to that of the RRKM calculation, and considering the difficulties inherent in the experiments, may be regarded as, at least, satisfactory. Since the thermal conductivities and enthalpies of isomerisation will be very similar for the two isotopic species, the interpretation of these isotope-ratio experiments is probably not seriously jeopardised by possible errors in the self-heating corrections, to which we referred above.

X. Future Developments

This reformulation of the theory of unimolecular reactions removes the need to invoke such troublesome artefacts as the 'activated complex', the 'reaction coordinate', and the 'reaction path degeneracy'. One might ask why it is that RRKM theory, with all these hypothetical constructs, could have been so successful? Its use of the generalised Lindemann form [27] instead of the generalised Lindemann form [46] would obviously not be a major cause of error; likewise the use of a $k(\epsilon)$ function rather than eq. 36 (based on the sum of states) rather than eq. 41 (based on the density of states) only has a minor effect, cf. Fig. 2. The basic reason why RRKM theory works so well is because, through the use of the entropy of activation to define the 'activated complex', the state

sum of the latter is increased from the state density of the molecule given by the harmonic oscillator approximation to a form more nearly like that given by the rotating anharmonic oscillator approximation (cf. Fig. 7a): this is usually achieved in conventional calculations by reducing the frequencies of some of the oscillators in the molecule (5). The fall-off behaviour of a reaction is predominantly determined by the distribution of $\rho(\epsilon)k(\epsilon)$, which varies as $G_v^+(\epsilon - \epsilon_0)$, the vibrational state sum of the activated complex in the RRKM formulation. The vibrational state density ρ_v and state sum G_v of a molecule of v vibrations vary roughly as ϵ^s and ϵ^{s+1} respectively where s is related to the number of low-frequency modes (see section VIII in text). On the other hand, the vibration-rotational state density ρ_{vr} varies as $\epsilon^{s+(3/2)}$. The lowering of some (typically a few) of the oscillator frequencies in the activated complex means that s for the complex will be greater than that of the molecule. Hence $(s_{\text{complex}} + 1) \simeq (s_{\text{molecule}} + \frac{3}{2})$ and the functional forms of G_v^+ and ρ_{vr} are comparable.

A major impediment, of course, to the application of this reformulation is the remarkable paucity of anharmonicity data for the vibrational motions of polyatomic molecules. It is obvious, then, that particularly in the consideration of the unimolecular break-up of photochemically excited states, of transient species, or of ionic species, the RRKM approach, which is here in its infancy (76), will not be rapidly superseded: the problem of estimating both vibration frequencies and their anharmonicities for such transient species will not readily be overcome.

In the immediate future, extension of this work can proceed in three well-defined directions. The first is towards the consideration of simpler molecules, where we expect that some of our cavalier assumptions will begin to break down. One can envisage at least four kinds of failure. For very simple molecules, e.g. H_2^+ or H_2 , the lifetimes of quasibound states with respect to predissociation appear to be entirely random as a function of $(\epsilon - \epsilon_0)$ (77, 78), and so we can expect that for very simple molecules, eq. 33 for $k(\epsilon)$ will become inadequate. Along with this, the density of states in such small molecules will be reduced, so that the states above threshold do not overlap, and therefore energy cannot be regarded as flowing freely throughout the molecule. For methyl isocyanide and cyclopropane, the predissociation widths of the states above ϵ_∞ implied by eq. 41 are respectively of the order 10^4 and 10^5 times the mean spacings between the levels, and the only qualm one needs to have about unlimited intramolecular energy exchange is the requirement to conserve angular

momentum. Even below threshold, the density of states is such that collision broadening causes them to overlap significantly, dependent of course on the complexity of the molecule, the nature and duration of the collision, and the pressure. All this free intramolecular exchange of energy must die away as the molecule becomes too simple: we will do no more than mention a small selection of a growing literature related to this problem (79–82). Thirdly, there is the assumption, implicit in our use of eq. 33 that rotational and vibrational energy are equally effective in bringing about predissociation; while the actual effect of rotationally-induced predissociation remains to be documented, one may note the possible important implication of rotation in the 'anomalous' Arrhenius factors of some reactions (14). Fourthly, it may become impermissible to lump together, as does eq. 42, with a single relaxation time, the rotational and the vibrational relaxations of these small molecules. Obviously, it is some or all of these effects, in concert, that lie at the cause of the difference in shock-wave dissociation behaviour between very simple molecules and more complex molecules (83).

A second possible extension might be to the problems of multiphoton induced dissociation and of laser separation of isotopic molecules (84, 85). Many of the ideas implicit in this approach, including several outlined in the preceding paragraph, can be carried across to these new problems (85, 86). Unfortunately, until very recently (87), molecules which undergo well-characterised thermal unimolecular decompositions have not been studied by these new methods. It would be interesting to see laser isotope separation experiments undertaken for one of the five molecules discussed in this paper: ethyl chloride is the obvious candidate, despite the fact that the thermal rate constants are the least reproducible of the five sets of data we have analysed.

The third obvious extension is to the discussion of chemical activation. The problem is a little more difficult here, and to follow our present approach would require a set of matrix elements a_{ij} rather like those of eq. 42, but having two non-degenerate eigenvalues. This $(n - 2)$ -fold degenerate formulation would also be useful for describing thermal reactions yielding two distinctly different products, and perhaps with distinctly different fall-off characteristics (88, 89).

Finally, there is the problem of completing the achievement of our original objective, that of putting the theory of thermal unimolecular reactions on a sound, modern molecular-dynamic footing. This requires us to understand why it is that the predissociation rate constants for individual states, i.e. the k_i , which almost certainly will have random values over

a small range of ϵ , can have a coarse-grain average which is reasonably well⁴ approximated by eq. 33. A start on this problem was made by Polanyi and Wigner (90) in 1928, but we are still far from a complete understanding of this particular facet. In this year of his death, it seems entirely appropriate to us to dedicate this paper to the memory of Michael Polanyi (1892–1977) who, among his many feats of intellect, was the father of modern molecular dynamics.

Acknowledgement

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⁴Very recently, Reddy and Berry (87) have studied the isomerisation of methyl isocyanide induced by intense laser radiation at 7265 Å; this corresponds to a value of $\epsilon = 39.3$ kcal mol⁻¹, i.e. 1.1 kcal mol⁻¹ above threshold (cf. Table 3). From the pressure dependence of the isomerisation rate, one can deduce that $k(\epsilon)$ at this energy is $\sim 1 \times 10^7$ s⁻¹, whereas eq. 33 (or 41) gives $k(\epsilon) = 1.0 \times 10^7$ s⁻¹ for the same energy.

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Appendix 1

Our choice of [33] for $k(\epsilon)$ is based on two arguments, one theoretical and one experimental. Firstly, for every molecule considered in this paper, the use of [33] in this reformulation gives a better representation of the shape of the fall-off curve than does the use of [36]: the latter formula always leads to a slightly flatter fall-off curve; an example of this is shown in Fig. 2 for cyclopropane. However, we do not place excessive weight on this coincidence, since it could be an artefact of our particular choice for the a_{ij} elements, eq. 42. Secondly, we conclude from the analysis of experimental data in this appendix that there is no experimental justification for preferring modified Arrhenius [35] over strict Arrhenius [34] rate laws for thermal unimolecular reactions.

A survey of the literature would indicate wide acceptance for the view that experimental rate constants probably have a more complex temperature dependence than is given by eq. 34; common variants are the modified Arrhenius form [35], or more complicated forms in which both the activation energy and pre-exponential factor are temperature-dependent (27, 28). It is, of course, extraordinarily difficult to distinguish among these various forms for any particular set of experimental rate constants because this would require accurate measurements of the rates over very wide ranges of temperature (27), and to our knowledge, only three unimolecular reactions have been so studied. The thermal decomposition of di-*tert*-butylperoxide has been studied over a range of temperature from 363–677 K (29–31): over this range, the rate constant changes by about ten orders of magnitude, from about 10^{-7} s^{-1} to about 10^3 s^{-1} , and there is no clear evidence that it cannot be represented by the strict Arrhenius form [34]. The

TABLE 2. Spectroscopic constants in cm^{-1}

Molecule	C_3H_6	C_4H_8	CH_3NC	CD_3NC	$\text{C}_2\text{H}_5\text{NC}$	$\text{C}_2\text{H}_5\text{Cl}$
Reference	55	91	92	92	93	94
ν_i ; $100x_i(g_i)$	739; 1.5 (2) 854; 1.5 869; 1.8 (2) 963; 1.8 1029; 2.0 (2) 1133; 2.0 1188; 0.0 1188; 2.0 (2) 1442; 2.5 (2) 1479; 0.0 3028; 5.3 (2) 3038; 8.0 3082; 4.3 (2) 3103; 2.9	199; 0.0 626; 1.5 737; 1.5 749; 1.5 (2) 901; 2.0 (2) 926; 0.0 999; 2.0 1005; 2.0 1224; 2.0 (2) 1225; 2.0 1234; 2.0 1257; 2.0 1260; 2.0 (3) 1452; 2.0 (2) 1454; 2.0 1469; 2.0 2887; 8.0 (2) 2905; 8.0 2945; 8.0 2965; 3.0 (2) 2974; 3.0 2987; 3.0	263; 2.0 (2) 945; 2.0 1129; 1.5 (2) 1429; 2.5 1467; 2.5 (2) 2166; 2.0 2966; 8.0 3014; 4.0 (2)	249; 1.9 (2) 877; 1.9 900; 1.2 (2) 1117; 1.2 1058; 1.8 (2) 2166; 2.0 2251; 6.0 2263; 3.0 (2)	226; 2.5 ^a 270; 2.0 ^a 466; 2.0 720; 1.5 801; 1.5 842; 3.0 1013; 3.0 1108; 1.5 1146; 2.0 1354; 1.5 1390; 2.5 1457; 2.5 1461; 2.5 (2) 2155; 2.5 2901; 8.0 2958; 8.0 2974; 3.0 2990; 3.0 3005; 3.0	336; 2.0 676; 3.0 785; 1.5 972; 1.5 972; 3.0 1080; 1.5 1244; 2.0 1287; 1.5 1383; 2.0 1452; 2.0 (3) 2890; 8.0 2946; 8.0 2983; 4.0 (2) 3012; 3.0
$B_{\parallel}$	0.670 (97)	0.355 (93)	0.335 (9)	0.30 ^b	0.161 ^c	0.174 ^d
B_z	0.435 ^b	0.226 (95)	5.22 (9)	2.61 ^b	0.926 (93)	1.05 (96)
ν_{torsion}					203	243
V					1239 ^f	1291 (98)
zpe^e					518	524
B_r^e					3326	45.74

^aFrequencies estimated by Maloney and Rabinovitch (10).^bEstimated.^cMean of $B_x = 0.170 \text{ cm}^{-1}$ and $B_y = 0.152 \text{ cm}^{-1}$ (93).^dMean of $B_x = 0.183 \text{ cm}^{-1}$ and $B_y = 0.165 \text{ cm}^{-1}$ (96).^e $zpe = \frac{1}{2}(V - \nu)$ where V is the barrier height and ν is the torsional frequency; the rotational constant is then $B_r = \nu^2/V$ (99).^fEstimated from barrier for $\text{C}_2\text{H}_5\text{CN} = 1819 \text{ cm}^{-1}$ (100), i.e. $V(\text{C}_2\text{H}_5\text{NC}) \approx V(\text{C}_2\text{H}_5\text{CN}) \times \nu(\text{C}_2\text{H}_5\text{NC})/\nu(\text{C}_2\text{H}_5\text{CN}) \approx 1819 \times 203/298 \text{ cm}^{-1}$.

thermal isomerisation of cycloheptatriene has been studied over the temperature range 600–1400 K (32, 33); again, the rate constant, ranging from about 10^{-4} s^{-1} to about 10^5 s^{-1} , is well represented by [34]. Likewise, the thermal isomerisation of methyl isocyanide has been studied over the range 393–593 K, with the rate constant changing by about seven orders of magnitude from about 10^{-8} s^{-1} to about 10^{-1} s^{-1} at the high pressures (34): in this case, the (extrapolated) infinite-pressure rate data were fitted by least squares to an equation of the form

$$[66] \quad \ln k_{\infty} = A + BT^{-1} + CT^{-2}$$

and then the quadratic term was discarded as not being statistically significant (35). On the other hand, there have been some indications, backed by theoretical arguments (36), that the rates of decomposition of cyclopropane and cyclobutane fall markedly below those predicted by eq. 34 at very high temperatures, i.e. about 1100 K (37, 38); however, recent evidence suggests that these results are incorrect and that the rate constants for these reactions can be represented by eq. 34 at least up to 1500 K (39). On the basis of these observations, taken together with

the fact that the observed fall-off behaviour is always more consistent with [33] and [34] than with [35] and [36], we propose, as far as *unimolecular* reactions are concerned, to proceed on the basis that until proved otherwise, infinite-pressure rate constants obey the strict Arrhenius form, eq. 34.

Appendix 2

The purpose of this appendix is to study the sensitivity of the state density of a molecule to variations in anharmonicity factor.

Figure 7 illustrates the sensitivity of the state density of methyl isocyanide for $\epsilon \geq 13\,000 \text{ cm}^{-1}$ to variations in anharmonicity factor: Fig. 7a examines variations in the anharmonicity factor for $\nu_8 = 263 \text{ cm}^{-1}$ and Fig. 7b for $\nu_1 = 2966 \text{ cm}^{-1}$. Figure 7a, curve (i) shows the function $\rho(\epsilon)$ calculated assuming that all the vibrations in the molecule are harmonic; curves (ii)–(v) are calculated assuming that all the oscillators are anharmonic (with the x_i as listed in Table 2) except for the bending mode ν_8 , for which x_8 values of 0 (harmonic), 0.01, 0.02, and 0.03 respectively are assumed. Three conclusions can be drawn from Fig. 7a. One, that all the curves are essentially

the same shape: for example $\rho(16\,000)/\rho(13\,000) \simeq 5.5$ for all curves, which implies that the calculated *shape* of the fall-off curve will be essentially the same for all five models; on the other hand, the absolute magnitude of $\rho(\epsilon)$ for model (i) differs from the rest by as much as a factor of 2, which corresponds to a factor of two difference in the absolute pressure shift (i.e. the collisional efficiency factor λ , or $\bar{\lambda}$). Two, that as the anharmonicity factor increases from zero, the state density first increases to a maximum and then decreases, as is to be expected for low-frequency modes in the interesting energy range. Three, that $\rho(\epsilon)$ only changes by about 10% for a change in x_8 from 0.01 to 0.03 (the estimated range), so that one can expect to derive reasonably meaningful results using estimated x_i values. In Fig. 7b, similar results are presented for the symmetric stretching mode $\nu_1 = 2966\text{ cm}^{-1}$. Curves (i) and (ii) show the function $\rho(\epsilon)$ for the anharmonic oscillator model (Table 2) but with $x_1 = 0$, and $x_1 = 0.08$ respectively; curves for $0 < x_1 < 0.08$ fall in between, and are omitted for clarity. Thus, for high-frequency modes, the state density is very insensitive indeed to the chosen values of x_i .

Finally, we examine in Fig. 8 the behaviour of $\rho(\epsilon)$ as x_i changes from 0 to 0.09 for the rocking mode ($\nu_8 = 854\text{ cm}^{-1}$) of cyclopropane to ensure that the conclusions just reached are equally valid at much higher energies; the $\rho(\epsilon)$ curve for the harmonic oscillator model is also shown for comparison. Again, all the curves have essentially the same shape: for example $\rho(27\,400)/\rho(22\,500)$ is about 16 for Morse oscillators and about 14 for harmonic oscillators. Also, the absolute magnitude of $\rho(\epsilon)$ changes by about 12% for a change in x_8 from 0 to 0.03, compared with a 10% change for methyl isocyanide as x_8 went from 0.01 to 0.03, despite the higher values of ϵ in this case: indeed, numerical examination shows that the sensitivity of $\rho(\epsilon)$ to x_i varies roughly as the ratio of ϵ/ν .

Appendix 3. Molecular Constants and Reaction-rate Parameters

Reliable vibrational assignments, in the harmonic-oscillator approximation, are virtually complete for all the molecules considered in this paper: the only exceptions are for ethyl isocyanide where Maloney and Rabinovitch (10) estimated $\nu_{13} = 226\text{ cm}^{-1}$ and $\nu_{20} = 270\text{ cm}^{-1}$, and for the torsional oscillations of both ethyl chloride and ethyl isocyanide which we have treated as free internal rotations with the rotational constants derived from the torsional frequency and barrier height (99), cf. Table 2.

The calculation of anharmonic force constants, on the other hand, has been largely restricted to tria-

tomic molecules, because of the enormous increase in the number of anharmonic terms in the potential function as the number of atoms increases (54). For unimolecular reaction systems, the most complete set of anharmonicity factors is that for cyclopropane, derived from approximate calculations by Cyvin (55). The anharmonicity factors x_i for the various modes are generally between 0.01 and 0.03, except for the stretching mode ($\nu_1 = 3038\text{ cm}^{-1}$) for which $x_1 \simeq 0.08$; to our knowledge, no complete set of anharmonicity factors exists for any other large molecule of interest in thermal unimolecular reactions. To define our notation, we denote the standard frequency assignments for the molecule by ν_i and the anharmonicity factors x_i ($\equiv$ the x^* of Cyvin) such that

$$[67] \quad \epsilon_{v,i} = \frac{\nu}{1 - x_i} [\nu_i - \nu \nu_i x_i]$$

Since such data are unlikely to be available in the near future, and in view of the mild insensitivity of $\rho(\epsilon)$ and the fall-off behaviour to the anharmonicity, we feel that, for the time being, it is a useful procedure to estimate values of x_i as follows, based on a comparison with Cyvin's (55) data for cyclopropane:

CH ₃ rocking	$x_i \simeq 0.015$
CH ₂ rocking	$x_i \simeq 0.015$
CH ₂ twisting	$x_i \simeq 0.02$
CH ₂ wagging	$x_i \simeq 0.02$
CH ₂ deformation	$x_i \simeq 0.025$
CH antisymmetric stretch	$x_i \simeq 0.03$
CH degenerate stretch	$x_i \simeq 0.04$
CH symmetric stretch	$x_i \simeq 0.08$
C—C stretch	$x_i \simeq 0.02$
C—N stretch	$x_i \simeq 0.02$
Bending modes	$x_i \simeq 0.02$

Since the CH stretching modes have high frequencies, of the order of 3000 cm^{-1} , the state density is not sensitive to their larger values of x_i in the interesting energy range. All the lower-frequency modes, which make the largest contributions to $\rho(\epsilon)$, appear to have values of x_i near 0.02, and in the absence of very refined calculations, this would appear to be about as close as one can get to the correct values. In the case of the isotopic pair of molecules CH₃NC and CD₃NC, we have assumed that the anharmonicity corrections are related by

$$[68] \quad \frac{\nu_i x_i(\text{D})}{\nu_i x_i(\text{H})} = \left[\frac{\nu_i(\text{D})}{\nu_i(\text{H})} \right]^2$$

Table 2 lists all pertinent spectroscopic information, observed or estimated, for the six molecular species treated in this paper, i.e. vibration frequencies ν_i , anharmonicity factors, listed as $100x_i$, degen-

eracies g_i (where they are different from unity), and external rotational constants B_z , $B_{||}$. Table 3 gives a complete self-consistent set of kinetic parameters, with reference to the source of each key entry.

Appendix 4. Algorithm for Calculating k_{uni} and Computer Program

An annotated program of the algorithm for calculating k_{uni} , written in FORTRAN, together with detailed explanations of formulae, sample input data, and specimen output, is lodged with the Depository of Unpublished Data.⁵

The algorithm includes the following key steps:

(i) The vibrational state-degeneracy $g_v(\epsilon)$ is computed from $\epsilon = 0$ to ϵ_m by direct count. This procedure is similar to the Stein-Rabinovitch (51) method, but differs from it in some important respects. Counting is performed only at energies where an energy state is present which results in an appreciable saving in time when ϵ_m is not high (as in the case here where ϵ_m is a few thousand wave numbers). This is because there are only a few states near $\epsilon = 0$, and one is therefore performing the counting and convolution only over these states, and not at all the energy grids, as in the Stein-Rabinovitch method. Also, the convolution is performed before round-off of the energy levels: this reduces to a minimum (with little additional programming effort) the propagation of round-off errors in the subsequent convolution with the rotational state-sum.

(ii) The rotational state-sum $G_r(\epsilon)$ is calculated up to ϵ_m , and convoluted to obtain the vibration-rotation state-sum $G_{vr}(\epsilon)$. Again, this is a conventional procedure, but the convolution is performed downwards towards $\epsilon = 0$ (as opposed to the conventional scheme of starting upwards from $\epsilon = 0$), resulting in a saving of half the required storage space.

(iii) If there are any internal rotational degrees of freedom, the state-degeneracy $g_i(\epsilon)$ of the internal rotation is convoluted with $G_{vr}(\epsilon)$ to obtain the sum of internal states $G(\epsilon)$ of the molecule. Again, this convolution procedure is performed downward.

(iv) The sum of states from $\epsilon = \epsilon_m + \Delta\epsilon$ to $\epsilon = \epsilon_{max}$ is calculated at intervals of $\Delta\epsilon$ using the non-iterative steepest-descent procedure (43).

(v) The density of states $\rho(\epsilon)$ is calculated from ϵ_{∞} to $\epsilon_{\infty} + \epsilon_{max}$ using the same procedure.

(vi) The reactive energy range (ϵ_{∞} to $\epsilon_{\infty} + \epsilon_{max}$) is grained into r channels, with states between ϵ_l and ϵ_{l+1} being grouped into the l th channel: the channels are not necessarily equally spaced; the values of $\bar{n}_l$

⁵Complete set of material is available upon request, at a nominal charge, from the Depository of Unpublished Data, CISTI, National Research Council of Canada, Ottawa, Ont., Canada K1A 0S2.

TABLE 3. Assumed values of the chemical kinetic rate parameters at infinite pressure: ϵ_{∞} in kcal mol⁻¹, A_{∞} and k_{∞} in s⁻¹^a

Molecule	C ₃ H ₆	C ₄ H ₈	CH ₃ NC ^b	CD ₃ NC	C ₂ H ₅ NC	C ₂ H ₅ Cl
Figure	9	10	12	19	13	11
ϵ_{∞} (kcal/mol)	65.6 (59)	62.5 (73)	38.4 (9)	38.5 (75)	38.4 (10)	58.4 (64)
A_{∞} (s ⁻¹)	1.88×10^{15}	3.66×10^{14}	3.98×10^{13} (9)	4.46×10^{13}	7.76×10^{13} (10)	1.07×10^{14} (64)
k_{∞} (s ⁻¹)	3.07×10^{-4} (58)	4.26×10^{-5} (58)	8.57×10^{-4}	8.65×10^{-4} (75)	1.73×10^{-3}	8.88×10^{-3}
At T (K)	763.15	721.45	503.55	503.55	504.05	794.15

^aReference numbers in parentheses.
^bThe literature set of values (34) used to construct Fig. 14 are $\epsilon_{\infty} = 38.2$ kcal mol⁻¹, $A_{\infty} = 2.24 \times 10^{13}$ s⁻¹, $k_{\infty} = 4.81 \times 10^{-5}$ s⁻¹ at 472.55 K, 5.89×10^{-4} s⁻¹ at 503.55 K, and 4.83×10^{-3} s⁻¹ at 532.95 K.

and $\bar{k}_i$ are calculated. The grouped decay rates $\bar{k}_i$ are scaled by the characteristic relaxation rate $\bar{\mu}$ (the relaxation rate at $p = 1$ Torr as defined in eq. 61) to avoid overflow or underflow difficulties in (vii) and (viii).

(vii) The scaled coefficients $\bar{a}_i$ and $\bar{b}_i$ for use in eq. 46 are computed using $\bar{n}_i$ and $\bar{k}_i$; ($\bar{k}_i = \bar{k}_i/\bar{\mu}$).

(viii) The unimolecular reaction rate constant k_{uni} may then be calculated for any value of the pressure p .

The FORTRAN program consists of the MAIN routine, which handles the flow of the program and the input data, and the following subroutines. (1) COUNT which makes a direct count of the vibrational energy level at low energy; (2) RINCNT which effects the convolution of any *internal* rotational degrees of freedom with the vibrational states; (3) EXTROT which brings about the convolution of the external rotational degrees of freedom; (4) DESENT which computes the state sum and state density of the molecule at higher energy by the non-iterative steepest-descent procedure (43); (5) GRAIN which divides the reactive energy region into grains and evaluates the quantities $\bar{n}_i$ and $\bar{k}_i$; (6) UNIRAT which computes the unimolecular rate constant at the de-

sired values of the pressure p ; (7) SETPLT (optional), which makes a Calcomp plot of the calculated rate constants as a function of pressure: this subroutine also reads in and plots specimen sets of experimental data, having determined the optimum value of the pressure-shift parameter $\bar{\lambda}$ required to effect the best coincidence between the calculated and experimental fall-off curves; (8) PFNCTN: a utility subroutine for calculating the molecular partition function; (9) FNPHI1 and FNPHI2: utility subroutines for the evaluation of the average internal energy and its first two derivatives, using exact and approximate (53) Morse partition functions respectively. All the desired options are specified in the input data set, and detailed descriptions of the input parameters and the structures of the respective subroutines are given as comments in the program listings.

The sample calculation evaluates k_{uni} for the methyl isocyanide isomerisation at about 50 values of p and matches the calculated curve with the experimental results of Collister and Pritchard (34) for the temperature $T = 230.4^\circ\text{C}$, i.e. the middle curve of Fig. 14; the complete calculation took about 20 s on the York University IBM 370/158 Computer.